



US012419139B2

(12) **United States Patent**
Hirano et al.

(10) **Patent No.:** **US 12,419,139 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **NITRIDE SEMICONDUCTOR ULTRAVIOLET LIGHT-EMITTING ELEMENT AND PRODUCTION METHOD THEREFOR**

(56) **References Cited**

U.S. PATENT DOCUMENTS

10,896,818 B2 1/2021 Han et al.
11,049,999 B2 6/2021 Hirano et al.
(Continued)

FOREIGN PATENT DOCUMENTS

JP 2002016000 A 1/2002
JP 2006060164 A 3/2006
(Continued)

OTHER PUBLICATIONS

Kojima et al., "Carrier localization structure combined with current micropaths in AlGa_N quantum wells grown on an AlN template with macrosteps," Applied Physics letter 114, Jan. 7, 2019, 6 pages.
(Continued)

Primary Examiner — Bradley Smith

(74) *Attorney, Agent, or Firm* — Andrew L. Dunlap;
Haynes Beffel & Wolfeld LLP

(57) **ABSTRACT**

A nitride semiconductor ultraviolet light-emitting element is provided. The element includes a light-emitting element structure part with an n-type layer, an active layer, and a p-type layer stacked vertically, which are made of AlGa_N-based semiconductors with wurtzite structure. The n-type layer has an n-type AlGa_N-based semiconductor, the active layer has well layers including an AlGa_N based semiconductor, and the p-type layer has a p-type AlGa_N-based semiconductor. Each semiconductor layer in the n-type and the active layers is an epitaxially grown layer having a surface on which multi-step terraces parallel to the (0001) plane are formed. The n-type layer has first Ga-rich regions which include n-type AlGa_N regions in which an AlGa_N composition ratio is an integer ratio of Al₁Ga₁N₂. The well layer includes a second Ga-rich region, which includes an AlGa_N region in which an AlGa_N composition ratio is an integer ratio of Al₁Ga₂N₃.

13 Claims, 19 Drawing Sheets

(71) Applicant: **NIKKISO CO., LTD.**, Tokyo (JP)

(72) Inventors: **Akira Hirano**, Aichi (JP); **Yosuke Nagasawa**, Nara (JP)

(73) Assignee: **NIKKISO CO., LTD.**, Tokyo (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 568 days.

(21) Appl. No.: **17/926,240**

(22) PCT Filed: **Jun. 24, 2020**

(86) PCT No.: **PCT/JP2020/024828**

§ 371 (c)(1),

(2) Date: **Nov. 18, 2022**

(87) PCT Pub. No.: **WO2021/260850**

PCT Pub. Date: **Dec. 30, 2021**

(65) **Prior Publication Data**

US 2023/0197889 A1 Jun. 22, 2023

(51) **Int. Cl.**

H01L 33/00 (2010.01)

H10H 20/01 (2025.01)

(Continued)

(52) **U.S. Cl.**

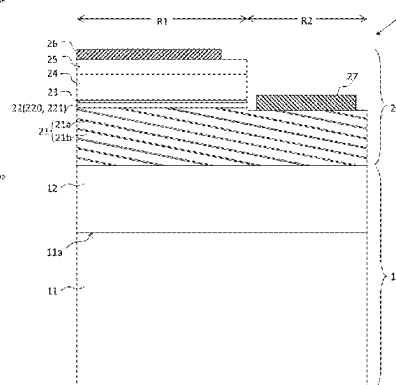
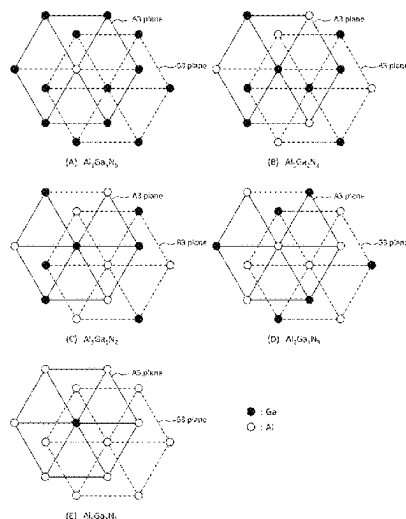
CPC **H10H 20/812** (2025.01); **H10H 20/01335** (2025.01); **H10H 20/817** (2025.01);

(Continued)

(58) **Field of Classification Search**

None

See application file for complete search history.



US 12,419,139 B2

Page 2

(51) **Int. Cl.**
H10H 20/812 (2025.01)
H10H 20/817 (2025.01)
H10H 20/818 (2025.01)
H10H 20/821 (2025.01)
H10H 20/825 (2025.01)
U.S. Cl.
CPC *H10H 20/818* (2025.01); *H10H 20/821* (2025.01); *H10H 20/825* (2025.01)

2018/0261725 A1 9/2018 Kaneda et al.
2020/0357953 A1 11/2020 Hirano et al.
2021/0043804 A1 2/2021 Hirano et al.

FOREIGN PATENT DOCUMENTS

JP 2016088803 A 5/2016
WO 2014178288 A1 11/2014
WO 2016157518 A1 10/2016
WO 2017013729 A1 1/2017
WO 2019102557 A1 5/2019
WO 2019159265 A1 8/2019

(56) References Cited

U.S. PATENT DOCUMENTS

11,152,543 B2 10/2021 Hirano et al.
2014/0158983 A1 * 6/2014 Pernot H10H 20/821
257/13
2015/0243856 A1 8/2015 Yamada et al.
2017/0033186 A1 * 2/2017 Han H01L 21/02433
2017/0263817 A1 9/2017 Hirano et al.

OTHER PUBLICATIONS

Nagasawa et al., "Comparison of Al_xGa_{1-x}N multiple quantum wells designed for 265 and 285nm deep-ultraviolet LEDs grown on AlN templates having macrosteps," Applied Physics Express 12, Jun. 4, 2019, 6 pages.

* cited by examiner

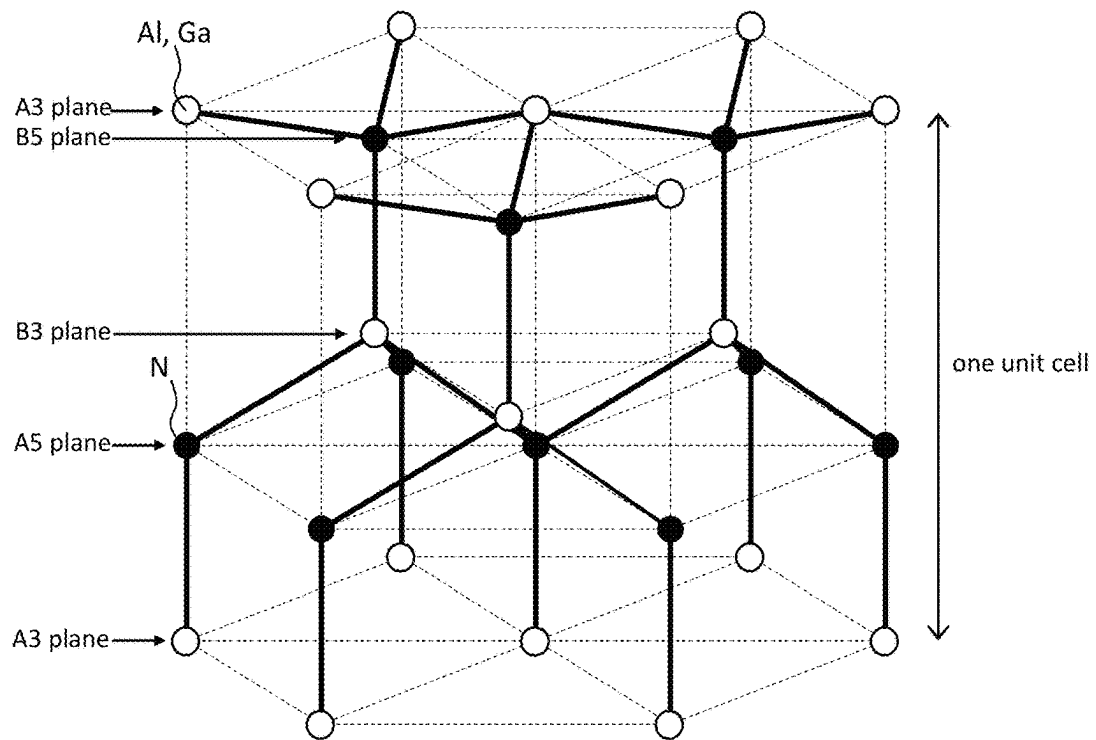


Fig. 1

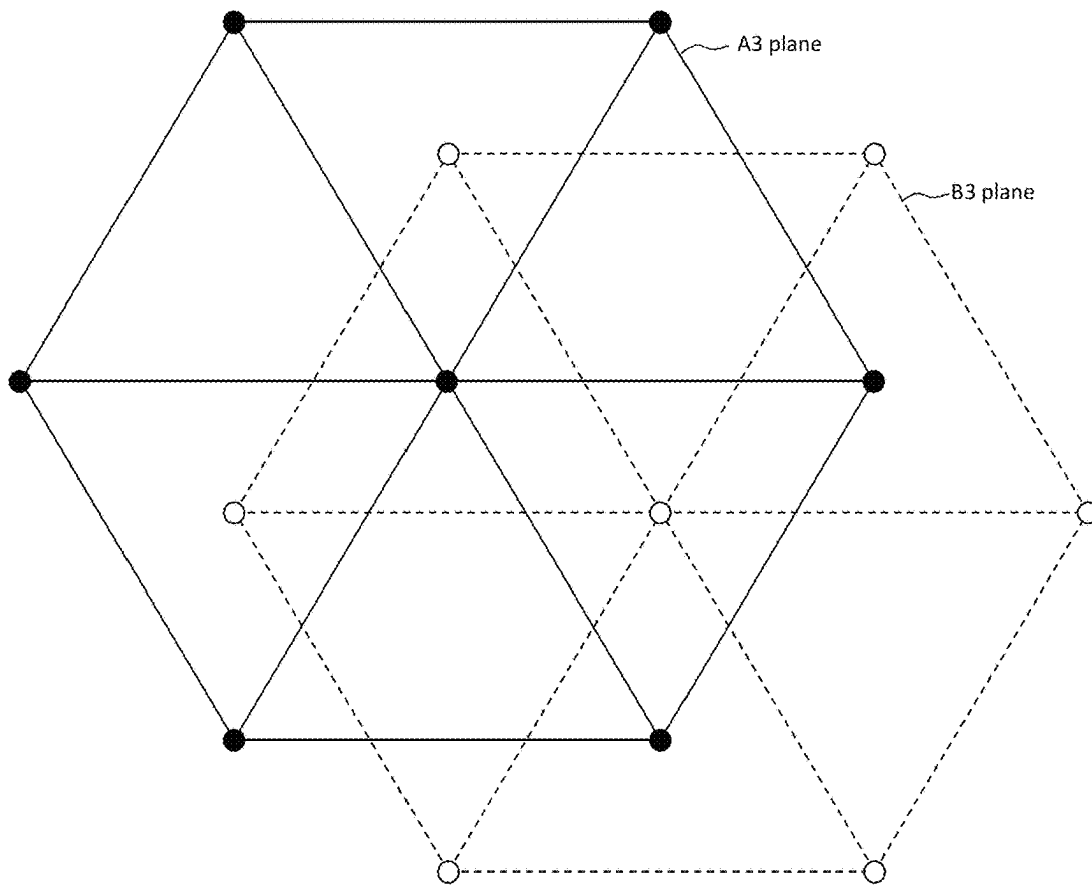


Fig. 2

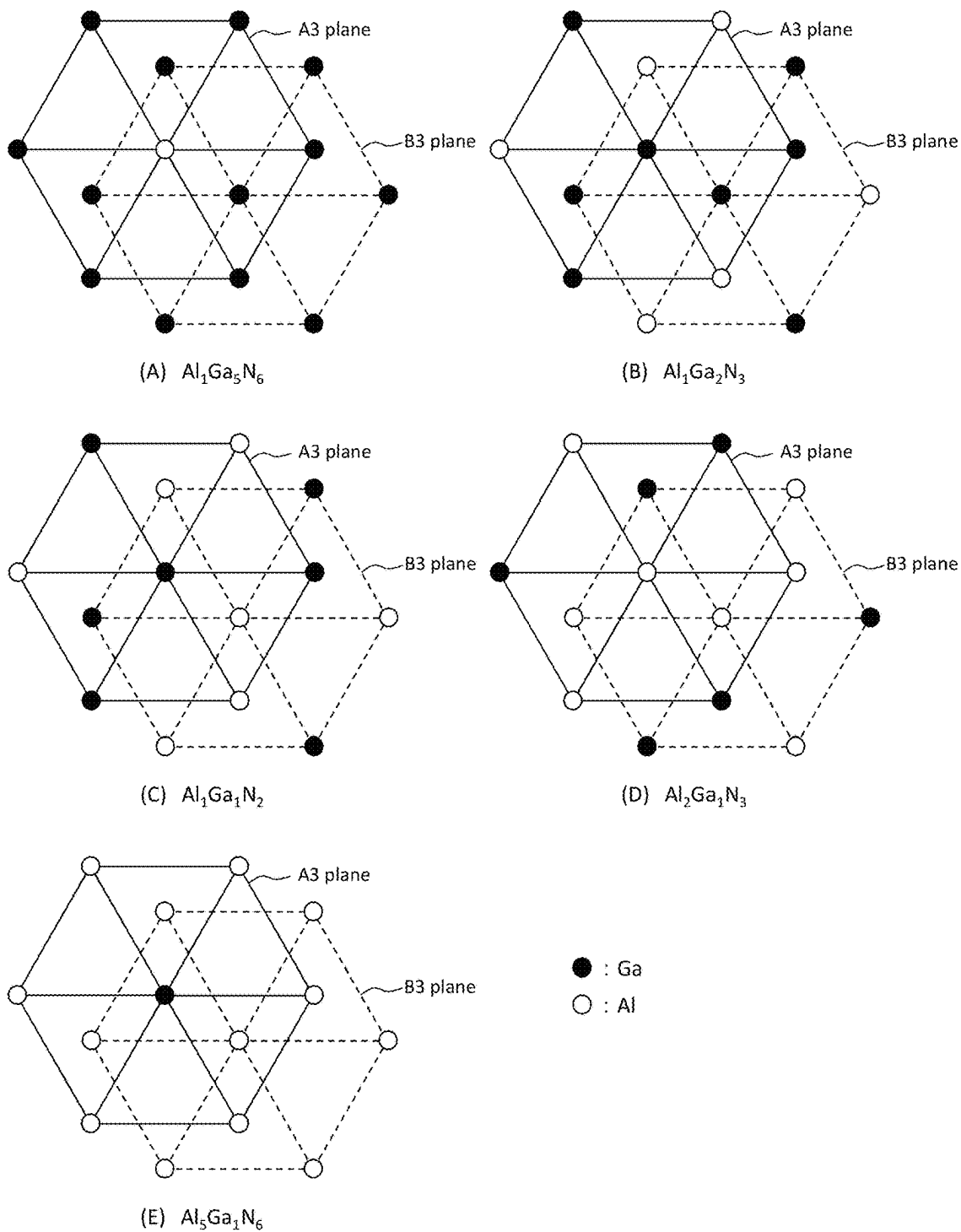


Fig. 3

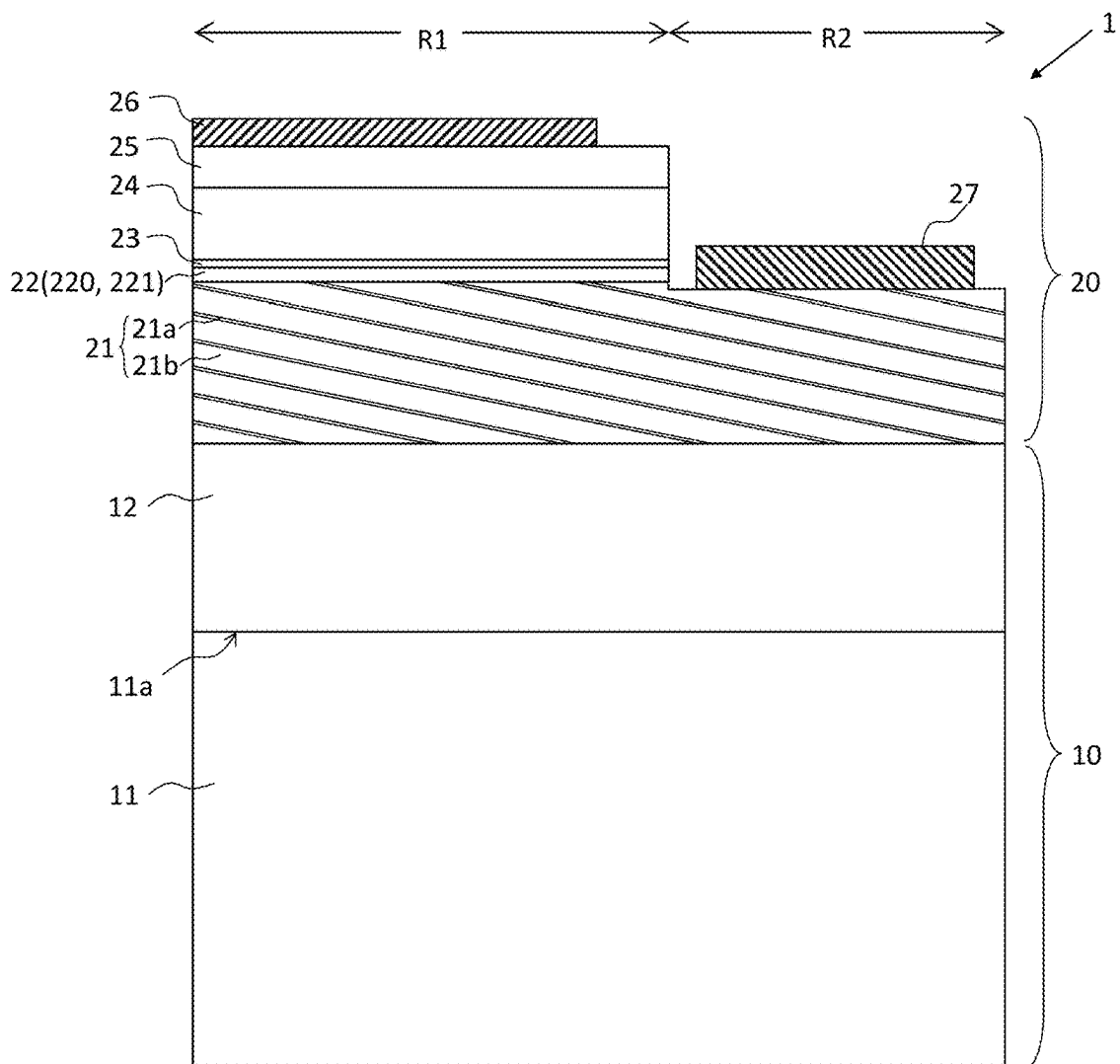


Fig. 4

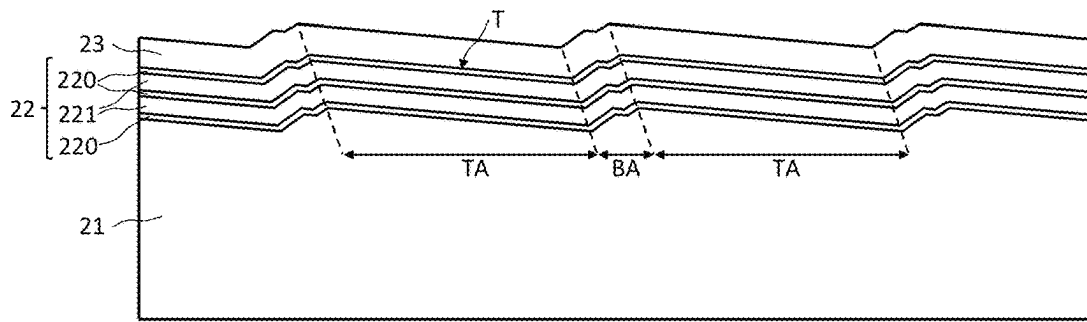


Fig. 5

Solid line: Barrier layer(66.7%) Dotted line: Barrier layer(75%) Dashed line: Barrier layer(83.3%)

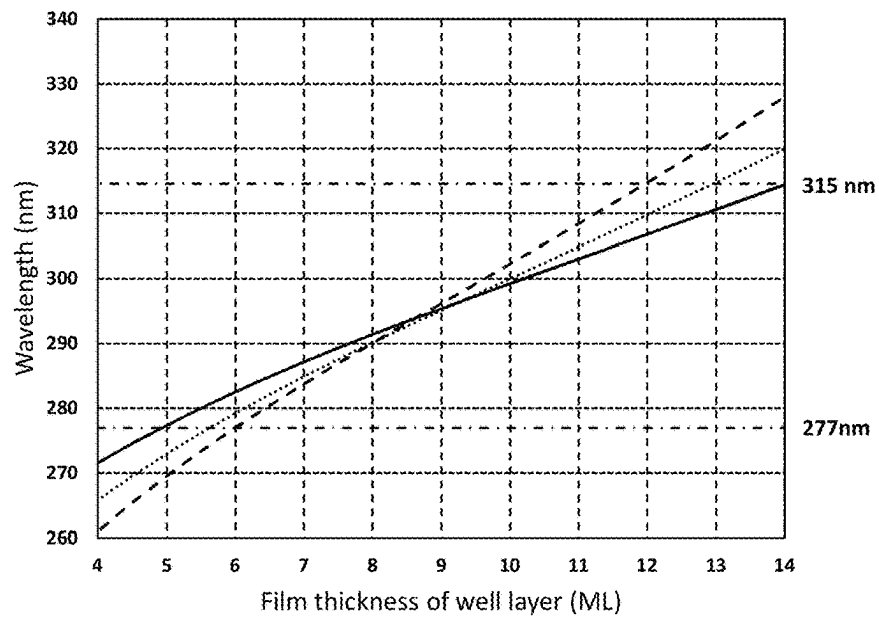


Fig. 6

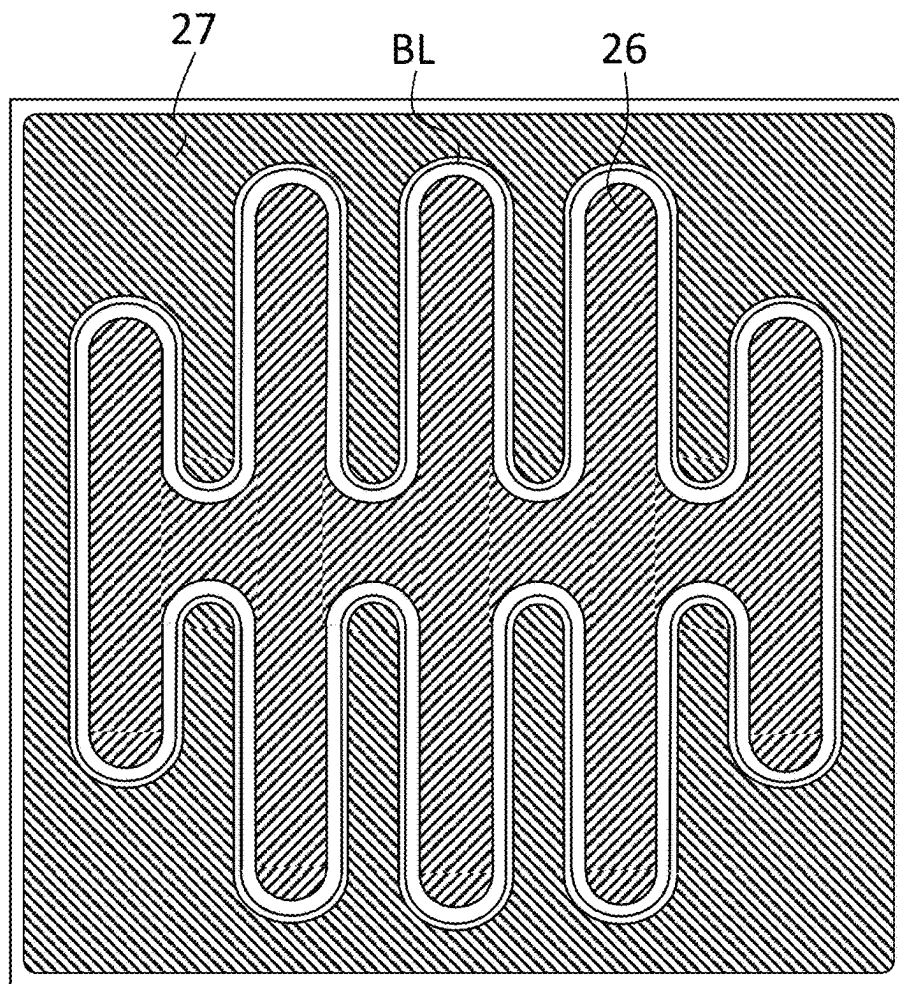


Fig. 7

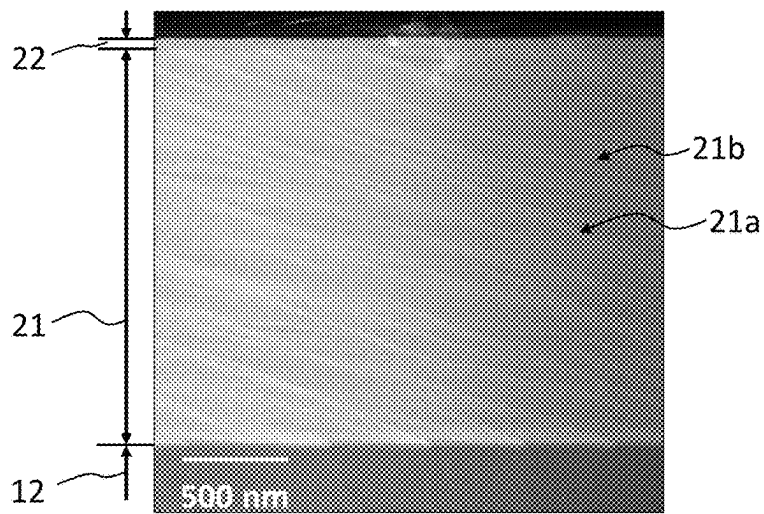


Fig. 8

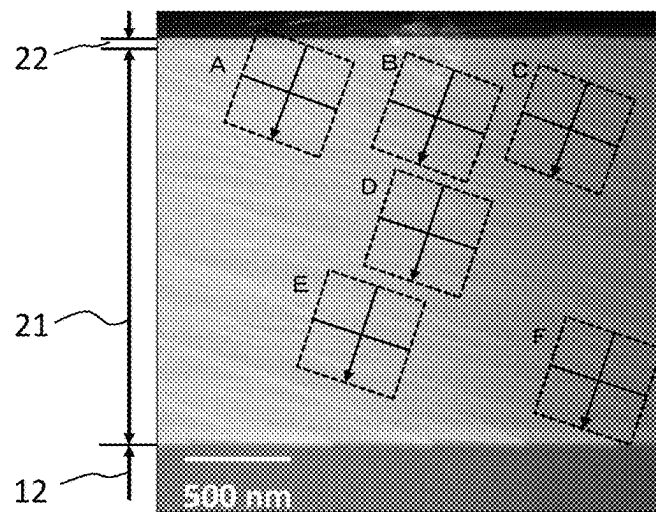


Fig. 9

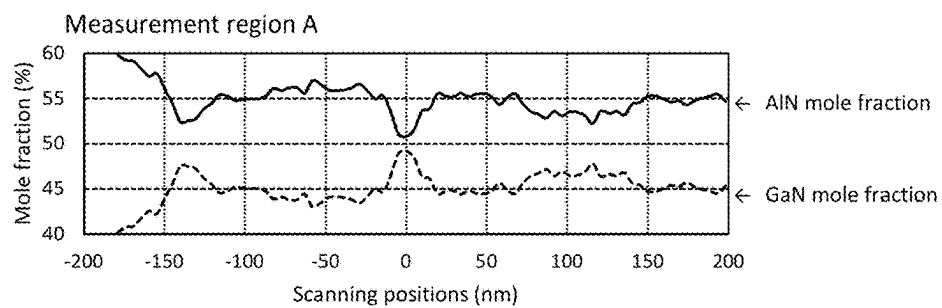


Fig. 10A

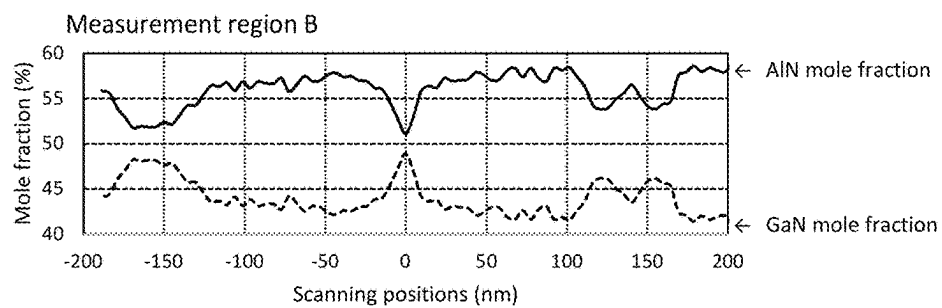


Fig. 10B

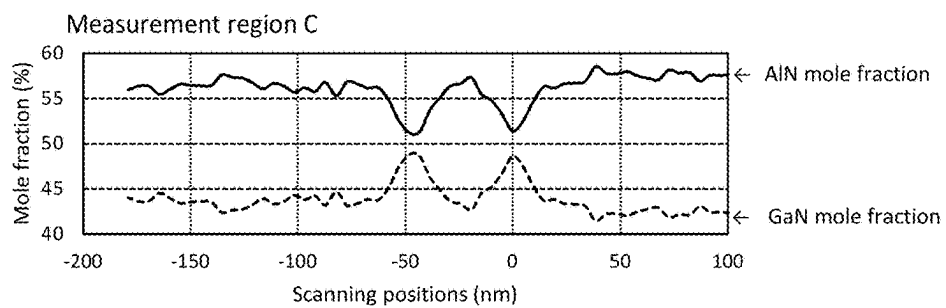


Fig. 10C

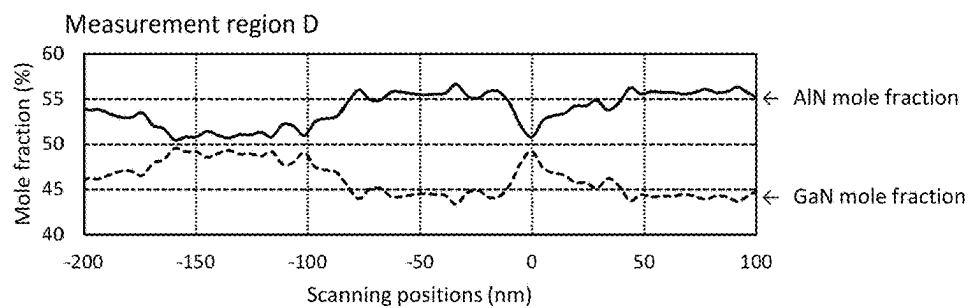


Fig. 10D

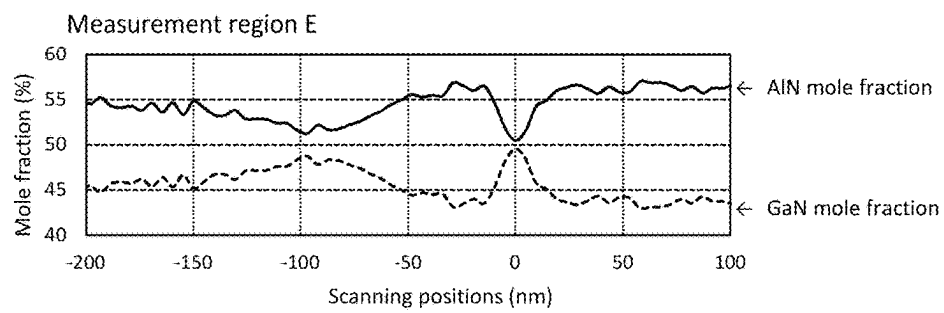


Fig. 10E

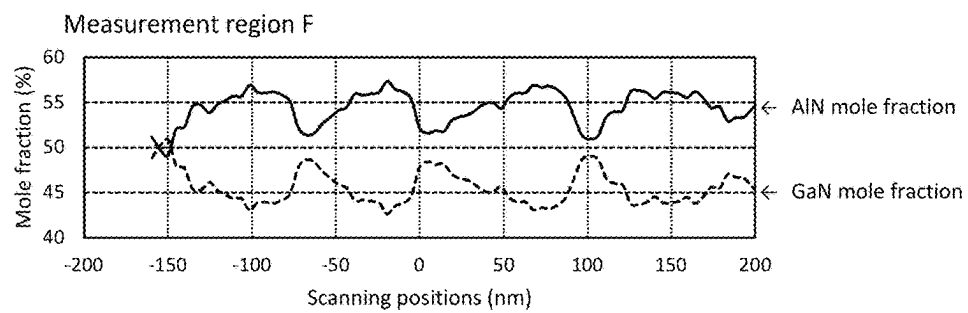


Fig. 10F

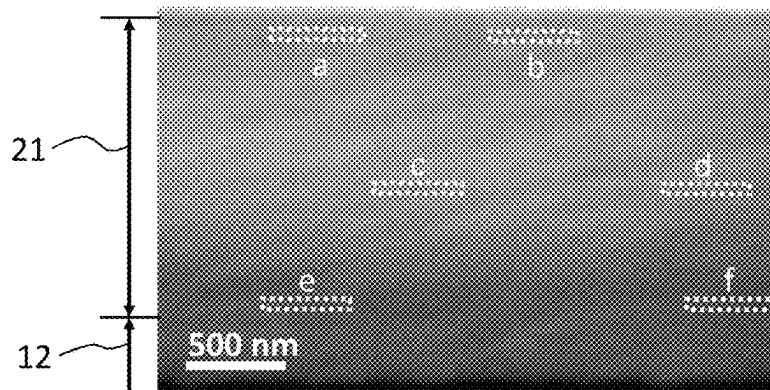


Fig. 11

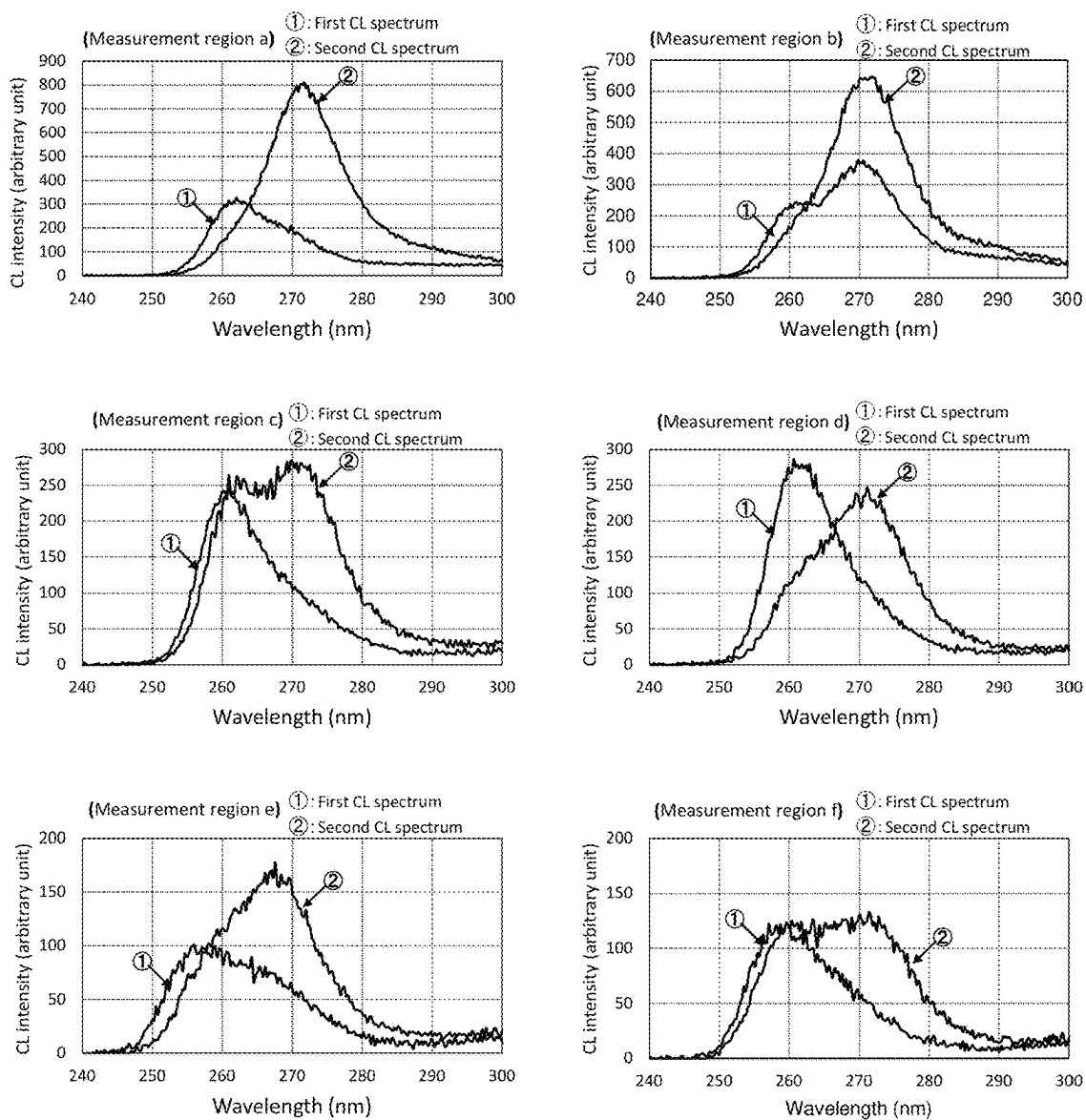


Fig. 12

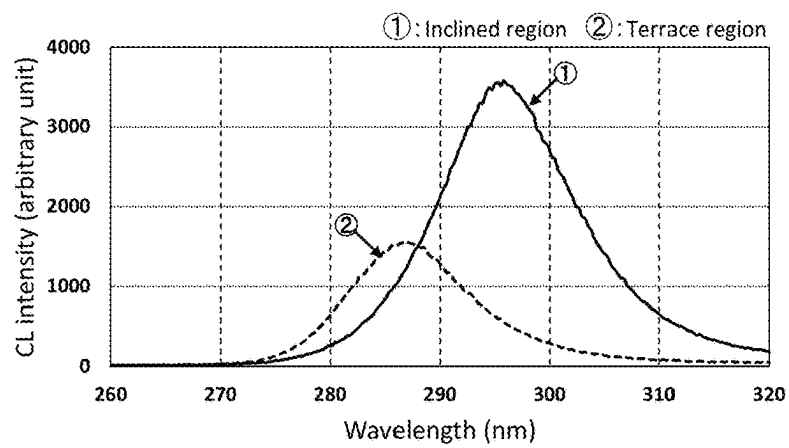


Fig. 13

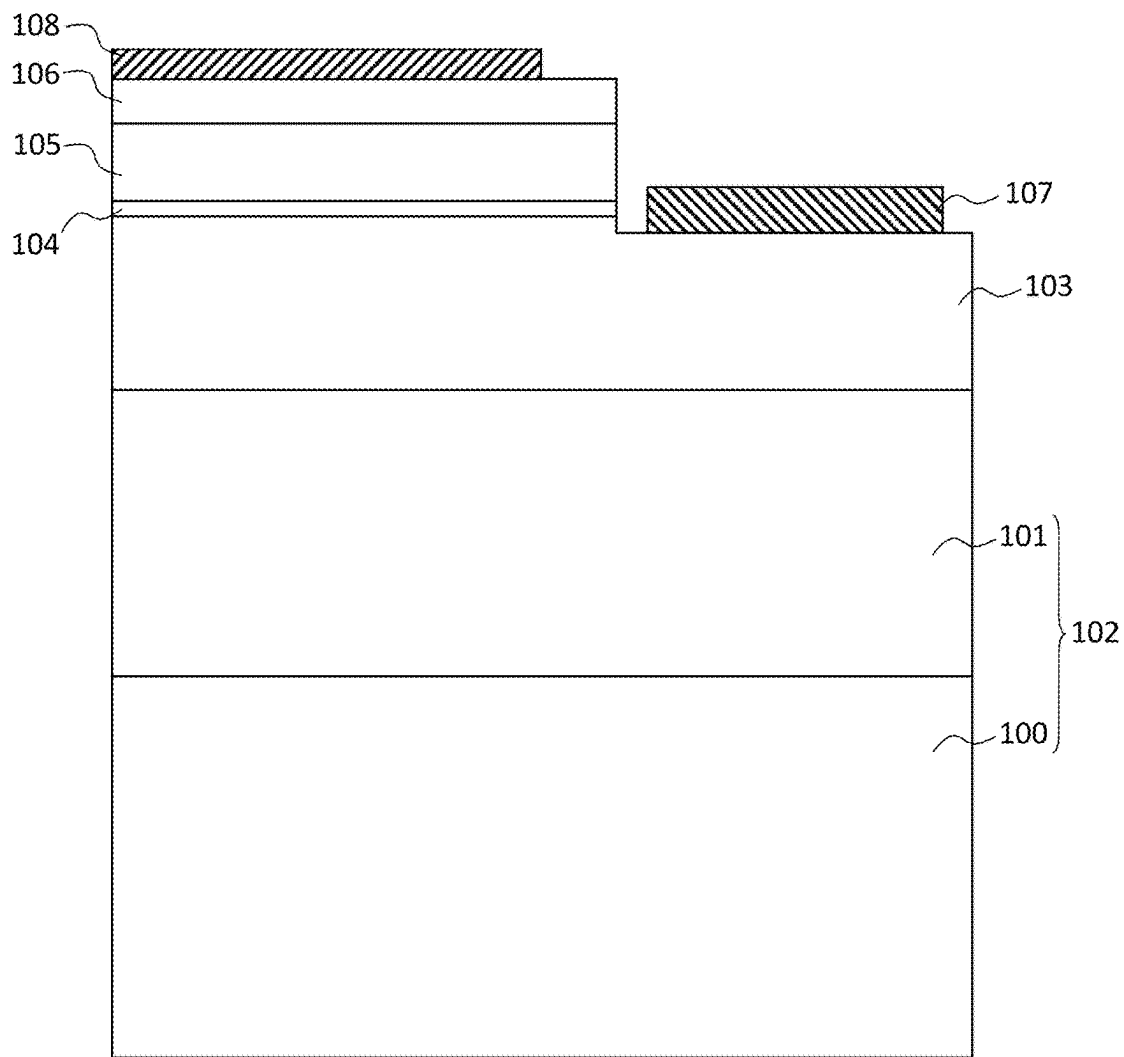


Fig. 14

1

NITRIDE SEMICONDUCTOR ULTRAVIOLET LIGHT-EMITTING ELEMENT AND PRODUCTION METHOD THEREFOR

TECHNICAL FIELD

The present invention relates to a nitride semiconductor ultraviolet light-emitting element comprising a light-emitting element structure part with an n-type layer, an active layer, and a p-type layer stacked vertically, which are made of AlGaIn-based semiconductors with wurtzite structure, and a manufacturing method thereof.

BACKGROUND ART

In general, there are a lot of nitride semiconductor light-emitting elements with a light-emitting element structure comprising a plurality of nitride semiconductor layers formed by epitaxial growth on a substrate such as sapphire. A nitride semiconductor layer is represented by the general formula: $\text{Al}_{1-x-y}\text{Ga}_x\text{In}_y\text{N}$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq x+y \leq 1$).

The light-emitting element structure of a light-emitting diode has a double hetero structure in which an active layer made of a nitride semiconductor layer having a single-quantum-well structure (SQW) or a multi-quantum-well structure (MQW) is sandwiched between two cladding layers of an n-type nitride semiconductor layer and a p-type nitride semiconductor layer. When the active layer is an AlGaIn-based semiconductor, by adjusting an AlN mole fraction (also referred to as an Al composition ratio), band gap energy can be adjusted within a range where band gap energies that can be taken by GaN and AlN (about 3.4 eV and about 6.2 eV) are lower and upper limits, so that an ultraviolet light-emitting element having a light emission wavelength of about 200 nm to about 365 nm is obtained. Specifically, by passing a forward current from the p-type nitride semiconductor layer toward the n-type nitride semiconductor layer, light emission corresponding to the band gap energy due to recombination of carriers (electrons and holes) occurs in the active layer. In order to supply the forward current from the outside, a p-electrode is provided on the p-type nitride semiconductor layer, and an n-electrode is provided on the n-type nitride semiconductor layer.

When the active layer is an AlGaIn-based semiconductor, the n-type nitride semiconductor layer and the p-type nitride semiconductor layer sandwiching the active layer are composed of the AlGaIn-based semiconductor having a higher AlN mole fraction than the active layer. However, since the p-type nitride semiconductor layer of a high AlN mole fraction is difficult to form a good ohmic contact with the p-electrode, it is generally performed to form a p-type contact layer made of a p-type AlGaIn semiconductor with a low AlN mole fraction (specifically p-GaN), which can have a good ohmic contact with the p-electrode, to the uppermost layer of the p-type nitride semiconductor layer. Since the AlN mole fraction of the p-type contact layer is smaller than that of the AlGaIn-based semiconductor constituting the active layer, ultraviolet light emitted toward the p-type nitride semiconductor layer side from the active layer is absorbed in the p-type contact layer and cannot be effectively extracted to the outside of the element. Therefore, a typical ultraviolet light-emitting diode having an active layer made of an AlGaIn-based semiconductor employs an element structure as schematically shown in FIG. 14. The ultraviolet light emitted toward the n-type nitride semiconductor layer side from the active layer is effectively

2

extracted to the outside of the element (e.g., see Patent Documents 1 and 2, Non-Patent Documents 1 and 2, etc. listed below).

As shown in FIG. 14, the typical UV light-emitting diode is constructed by depositing an n-type AlGaIn-based semiconductor layer 103, an active layer 104, a p-type AlGaIn-based semiconductor layer 105, and a p-type contact layer 106 on a template 102 formed by depositing an AlGaIn-based semiconductor layer 101 (e.g., AlN layer) on a substrate 100 such as a sapphire substrate, and etching away respective portions of the active layer 104, the p-type AlGaIn-based semiconductor layer 105, and the p-type contact layer 106 until the n-type AlGaIn-based semiconductor layer 103 is exposed, and forming an n-electrode 107 on the exposed surface of the n-type AlGaIn-based semiconductor layer 103 and an p-electrode 108 on the surface of the p-type contact layer 106.

In addition, in order to improve luminous efficiency (internal quantum efficiency) by carrier recombination in the active layer, the active layer is formed in a multi-quantum-well structure, and an electron blocking layer is provided on the active layer.

On the other hand, it has been reported that in a cladding layer made of an n-type AlGaIn-based semiconductor layer, compositional modulation due to segregation of Ga occurs, and stratiform regions with a locally lower AlN mole fraction are formed extending obliquely to the surface of the cladding layer (for example, see Patent Document 3, Non-Patent Documents 1, 2, etc. described below). Because the band gap energy of the AlGaIn-based semiconductor layer with the locally lower AlN mole fraction is also locally reduced, it is reported in Patent Document 3 that the carriers in the cladding layer is easily localized in the stratiform regions, which can provide low resistance current paths to the active layer, and that the luminous efficiency of the ultraviolet light-emitting diode can be improved.

PRIOR ART DOCUMENTS

Patent Document

Patent Document 1: WO2014/178288
Patent Document 2: WO2016/157518
Patent Document 3: WO2019/159265

Non-Patent Document

NON-PATENT DOCUMENT 1: Y. Nagasawa, et al., "Comparison of $\text{Al}_x\text{Ga}_{1-x}\text{N}$ multiple quantum wells designed for 265 and 285 nm deep-ultraviolet LEDs grown on AlN templates having macrosteps", Applied Physics Express 12, 064009 (2019)

NON-PATENT DOCUMENT 2: K. Kojima, et al., "Carrier localization structure combined with current micropaths in AlGaIn quantum wells grown on an AlN template with macrosteps", Applied Physics letter 114, 011102 (2019)

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

An ultraviolet light-emitting element composed of the AlGaIn-based semiconductors is produced on a substrate such as a sapphire substrate by a well-known epitaxial growth method such as, for example, a metalorganic vapor phase epitaxy (MOVPE) method. However, when producing the ultraviolet light-emitting element, the characteristics of

3

the ultraviolet light-emitting element (properties such as emission wavelength, wall plug efficiency, forward bias, etc.) fluctuate under the influence by drift of the crystal growth apparatus, so that the producing at a stable yield is not always easy.

The drift of the crystal growth apparatus occurs due to a change in the effective temperature of the crystal growth site or the like owing to deposits on trays, chamber walls, or the like. Therefore, in order to suppress the drift, conventionally, treatments such as fine adjustments of the set temperature and the composition of source gases by experienced persons with examining the growth history, fixing the growth schedule for a certain period, and performing maintenance such as cleaning in the same manner for a certain period are performed, but it is difficult to eliminate the drift completely.

The present invention has been made in view of the above-mentioned problems, and an object of the present invention is to provide a nitride semiconductor ultraviolet light-emitting element which can be stably produced with variation in characteristics due to the drift of the crystal growth apparatus or the like being suppressed.

Means for Solving the Problem

The present invention, in order to achieve the above object, provides a nitride semiconductor ultraviolet light-emitting element comprising a light-emitting element structure part in which an n-type layer, an active layer, and a p-type layer made of an AlGaIn-based semiconductor of wurtzite structure are stacked vertically, wherein

the n-type layer is composed of an n-type AlGaIn-based semiconductor,

the active layer disposed between the n-type layer and the p-type layer has a quantum-well structure having one or more well layers composed of an AlGaIn-based semiconductor,

the p-type layer is composed of a p-type AlGaIn-based semiconductor,

each semiconductor layer in the n-type layer and the active layer is an epitaxially grown layer having a surface on which multi-step terraces parallel to a (0001) plane are formed,

the n-type layer has a plurality of first Ga-rich regions, the plurality of first Ga-rich regions being stratiform regions uniformly distributed in the n-type layer with locally lower AlN mole fraction and including n-type AlGaIn regions in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_1\text{N}_2$,

each extending direction of the first Ga-rich regions on a first plane perpendicular to an upper surface of the n-type layer is inclined with respect to an intersection line between the upper surface of the n-type layer and the first plane,

boundary region parts between adjacent terraces of the multi-step terraces of the well layer have a second Ga-rich region with locally lower AlN mole fraction in the same well layer, and

the second Ga-rich region includes an AlGaIn region in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_2\text{N}_3$.

Furthermore, the present invention, in order to achieve the above object, provides a manufacturing method of a nitride semiconductor ultraviolet light-emitting element comprising a light-emitting element structure part in which an n-type layer, an active layer, and a p-type layer made of an AlGaIn-based semiconductor of wurtzite structure are stacked vertically, the method comprising

4

a first step of epitaxially growing the n-type layer of an n-type AlGaIn-based semiconductor on an underlying part including a sapphire substrate having a main surface inclined by a predetermined angle with respect to a (0001) plane, and making multi-step terraces parallel to the (0001) plane appear on a surface of the n-type layer,

a second step of epitaxially growing the active layer of a quantum-well structure having one or more well layers composed of an AlGaIn-based semiconductor on the n-type layer, and making multi-step terraces parallel to the (0001) plane appear on a surface of the well layer, and

a third step of forming the p-type layer of a p-type AlGaIn-based semiconductor on the active layer by epitaxial growth, wherein

in the first step, a plurality of first Ga-rich regions are grown so as to extend obliquely upward, the plurality of first Ga-rich regions being stratiform regions uniformly distributed in the n-type layer with locally lower AlN mole fraction and including n-type AlGaIn regions in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_1\text{N}_2$, and

in the second step, a second Ga-rich region with locally lower AlN mole fraction in the same well layer is formed in boundary region parts between adjacent terraces of the multi-step terraces of the well layer, and an AlGaIn region in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_2\text{N}_3$ is grown in the second Ga-rich region.

The AlGaIn-based semiconductor is represented by the general formula $\text{Al}_{1-x}\text{Ga}_x\text{N}$ ($0 \leq x \leq 1$), but the semiconductor may contain a trace amount of an impurity such as a Group 3 element such as B or In or a Group 5 element such as P, as far as the band gap energy is within the lower limit and the upper limit of the band gap energy that can be obtained by GaN and AlN, respectively. The GaN-based semiconductor is basically a nitride semiconductor composed of Ga and N, and may contain a trace amount of an impurity such as a Group 3 element such as Al, B, or In or a Group 5 element such as P. The AlN-based semiconductor is basically a nitride semiconductor composed of Al and N, and may contain a trace amount of an impurity such as a Group 3 element such as Ga, B, or In or a Group 5 element such as P. Therefore, in the present application, the GaN-based semiconductor and the AlN-based semiconductor are a part of the AlGaIn-based semiconductor, respectively.

Furthermore, the n-type or p-type AlGaIn-based semiconductor is an AlGaIn-based semiconductor in which Si or Mg or the like is doped as a donor impurity or an acceptor impurity. In the present application, the AlGaIn-based semiconductor, not specified as p-type or n-type, means undoped AlGaIn-based semiconductor, but even if undoped, a trace amount of donor or acceptor impurities to the extent of being inevitably mixed may be included. The first plane is not an exposed surface or a boundary surface between the n-type layer and other semiconductor layer which is specifically formed in the manufacturing process of the n-type layer but is a virtual plane extending in parallel to the vertical direction in the n-type layer. Furthermore, in this specification, an AlGaIn-based semiconductor layer, a GaN-based semiconductor layer, and an AlN-based semiconductor layer are semiconductor layers composed of the AlGaIn-based semiconductor, the GaN-based semiconductor, and the AlN-based semiconductor, respectively.

According to the nitride semiconductor ultraviolet light-emitting element having the above-mentioned features or the

5

manufacturing method of the nitride semiconductor ultra-violet light-emitting element having the above-mentioned features, it is expected that a nitride semiconductor ultra-violet light-emitting element having the desired emission characteristic is stably produced by utilizing metastable AlGaN_s formed in the first Ga-rich region in the n-type layer and the second Ga-rich region in the well layer as described later to suppress the characteristic variation caused by the drift of the crystal growth apparatus or the like.

First, a “metastable AlGaN” in which an AlGaN composition ratio is represented by a predetermined integer ratio will be described.

Usually, a ternary mixed crystal such as AlGaN is a crystalline state in which group 3 elements (Al and Ga) are randomly mixed and is approximately described by “random non-uniformity”. However, because the covalent bond radius of Al and the covalent bond radius of Ga are different, the higher the symmetry of the atomic arrangement of Al and Ga in the crystal structure, the more stable the structure is in general.

The AlGaN-based semiconductors with wurtzite structure can have two types of arrangements, a random arrangement without symmetry and a stable symmetric arrangement. Here, a state in which the symmetric arrangement is dominant appears at a constant rate. In the “metastable AlGaN” described later, in which an AlGaN composition ratio (composition ratio of Al, Ga, and N) is represented by a predetermined integer ratio, a periodic symmetric arrangement structure of Al and Ga develops.

In the periodic symmetric arrangement structure, even if amount of Ga supplied to the crystal growth surface is slightly increased or decreased, the mixed crystal mole fraction becomes slightly stable in terms of energy because of the high symmetry, and it is possible to prevent proliferation of places where amount of easily mass-transferring Ga is extremely increased. That is, by utilizing the properties of the “metastable AlGaN” formed in the first Ga-rich region in the n-type layer, as an AlGaN-based semiconductor, even if the variation of the mixed crystal mole fraction due to the drift of the crystal growth apparatus or the like occurs, the variation of the mixed crystal mole fraction in the first Ga-rich regions which provide the low resistance current paths to the active layer as described later is locally suppressed. Consequently, stable carrier supply from the n-type layer into the active layer can be realized and variation in device characteristics can be suppressed, so that it is expected to stably produce a nitride semiconductor ultra-violet light-emitting element that achieves the desired characteristics.

Next, the AlGaN composition ratio in which Al and Ga can be in the periodic symmetric arrangement in the (0001) plane will be described.

FIG. 1 shows a schematic diagram of one unit cell (two monolayers) along the c-axis of AlGaN. In FIG. 1, open circles indicate sites where atoms of Group 3 elements (Al, Ga) are located, and solid circles indicate sites where atoms of Group 5 elements (N) are located.

The site planes (A3 plane, B3 plane) of the Group 3 elements and the site planes (A5 plane, B5 plane) of the Group 5 element shown by hexagons in FIG. 1 are both parallel to the (0001) plane. Six sites at each vertex of the hexagon and one site at the center of the hexagon are present at each site of the A3 and A5 planes (collectively, plane A). The same applies to the B3 plane and the B5 plane (collectively B plane), but in FIG. 1, it illustrates only three sites present in the hexagon of the B plane. Each site of the A plane is overlapped in the c-axis direction, each site of the

6

B plane is overlapped in the c-axis direction. However, the atom (N) of one site on the B5 plane forms a quaternary bond with the atoms (Al, Ga) of the three sites on the A3 plane located above the B5 plane and the atom (Al, Ga) of one site on the B3 plane located below the B5 plane, and the atom (Al, Ga) of one site on the B3 plane forms a quaternary bond with the atom (N) of one site on the B5 plane located above the B3 plane and the atoms (N) of three sites on the A5 plane located below the B3 plane, so that each site on the A plane does not overlap each site on the B plane in the c-axis direction, as shown in FIG. 1.

FIG. 2 shows a positional relationship between each site of the A plane and the B plane, as a plan view as viewed from the c-axis direction. In both the A and B planes, each of the six vertices of the hexagon is shared by the other two hexagons adjacent to each other, and the site at the center is not shared with the other hexagons, so there are substantially three atomic sites within one hexagon. Thus, there are six sites of Group 3 element atoms (Al, Ga) and six sites of Group 5 element atoms (N) per unit cell. Therefore, the following five cases exist as AlGaN composition ratios expressed by the integer ratio excluding GaN and AlN.

- 1) $\text{Al}_1\text{Ga}_5\text{N}_6$
- 2) $\text{Al}_2\text{Ga}_4\text{N}_6 (= \text{Al}_1\text{Ga}_2\text{N}_3)$
- 3) $\text{Al}_3\text{Ga}_3\text{N}_6 (= \text{Al}_1\text{Ga}_1\text{N}_2)$
- 4) $\text{Al}_4\text{Ga}_2\text{N}_6 (= \text{Al}_2\text{Ga}_1\text{N}_3)$
- 5) $\text{Al}_5\text{Ga}_1\text{N}_6$

FIG. 3 schematically shows the A3 plane and the B3 plane of the group 3 element in the above five combinations. Ga is indicated by a solid circle, and Al is indicated by an open circle.

In the case of $\text{Al}_1\text{Ga}_5\text{N}_6$ shown in FIG. 3(A), Ga is located at six vertex sites of the A3 plane and six vertex sites and one center site of the B3 plane, and Al is located at one center site of the A3 plane.

In the case of $\text{Al}_1\text{Ga}_2\text{N}_3$ shown in FIG. 3(B), Ga is located at three vertex sites and one center site of the A3 and B3 planes, and Al is located at three vertex sites of the A3 and B3 planes.

In the case of $\text{Al}_1\text{Ga}_1\text{N}_2$ shown in FIG. 3(C), Ga is located at three vertex sites and one center site of the A3 plane and three vertex sites of the B3 plane, and Al is located at three vertex sites of the A3 plane and three vertex sites and one center site of the B3 plane.

In the case of $\text{Al}_2\text{Ga}_1\text{N}_3$ shown in FIG. 3(D), Ga is located at three vertex sites of the A3 and B3 planes, Al is located at three vertex sites and one center site of the A3 and B3 planes. This is equivalent to swapping the positions of Al and Ga in $\text{Al}_1\text{Ga}_2\text{N}_3$ shown in FIG. 3(B).

In the case of $\text{Al}_5\text{Ga}_1\text{N}_6$ shown in FIG. 3(E), Ga is located at one center site of the A3 plane, Al is located at six vertex sites of the A3 plane and six vertex sites and one center site of the B3 plane. This is equivalent to swapping the positions of Al and Ga in $\text{Al}_1\text{Ga}_5\text{N}_6$ shown in FIG. 3(A).

In each of FIGS. 3(A)-3(E), assuming another hexagon whose center has moved to any one of the six vertices of the hexagon, it can be seen that Al or Ga located at the six vertex sites of the A3 plane is equivalent to Al or Ga located at the three vertex sites and the one center site of the A3 plane, and Al or Ga located at the one center of the A3 plane is equivalent to Al or Ga located at the three vertex sites of the A3 plane. The same applies to the B3 plane. In addition, in each of FIGS. 3(A), 3(C) and 3(E), A3 and B3 planes may be replaced with each other.

In each of FIGS. 3(A)-3(E), in both A3 and B3 planes, the atomic arrangement of Al and Ga is maintained in symmetry.

Even if the center of the hexagon is moved, the atomic arrangement of Al and Ga is maintained in symmetry

Furthermore, in the A3 and B3 planes of FIGS. 3(A)-3(E), when the hexagonal site plane is arranged repeatedly in a honeycomb shape, looking at each site in a direction parallel to the (0001) plane, for example, in [11-20] direction or [10-10] direction, the state in which Al and Ga is located periodically repeatedly or either Al or Ga is located continuously appears. Therefore, it can be seen that an atomic arrangement will be the periodic and symmetric atomic arrangement in the respective cases.

Hereinafter, $\text{Al}_{1-x_1}\text{Ga}_{x_1}\text{N}$ of the AlN mole fraction x_1 ($x_1=1/6, 1/3, 1/2, 2/3, 5/6$) corresponding to AlGa₃N composition ratios of above-mentioned 1) to 5) is referred to as the “first metastable AlGa₃N”, for convenience of explanation. In the first metastable AlGa₃N, the atomic arrangement of Al and Ga becomes a periodic and symmetric arrangement, resulting in an energetically stable AlGa₃N.

Next, when the site plane indicated by the hexagon shown in FIG. 1 is extended to 2 unit cells (4 monolayers), there are two planes for each of the site planes of the Group 3 element (A3 plane, B3 plane) and two planes for each of the site planes of the Group 5 element (A5 plane, B5 plane), respectively, and there are 12 sites for atoms of the Group 3 element (Al, Ga) and 12 sites for atoms of the Group 5 element (N) per 2 unit cells. Therefore, as AlGa₃N composition ratios expressed by the integer ratio excluding GaN and AlN, the following six combinations exist in addition to the AlGa₃N composition ratios of above-mentioned 1) to 5).

- 6) $\text{Al}_1\text{Ga}_{11}\text{N}_{12}$ ($=\text{GaN}+\text{Al}_1\text{Ga}_5\text{N}_6$)
- 7) $\text{Al}_3\text{Ga}_9\text{N}_{12}$ ($=\text{Al}_1\text{Ga}_3\text{N}_4+\text{Al}_1\text{Ga}_5\text{N}_6+\text{Al}_1\text{Ga}_2\text{N}_3$)
- 8) $\text{Al}_5\text{Ga}_7\text{N}_{12}$ ($=\text{Al}_1\text{Ga}_2\text{N}_3+\text{Al}_1\text{Ga}_1\text{N}_2$)
- 9) $\text{Al}_7\text{Ga}_5\text{N}_{12}$ ($=\text{Al}_1\text{Ga}_1\text{N}_2+\text{Al}_2\text{Ga}_1\text{N}_3$)
- 10) $\text{Al}_9\text{Ga}_3\text{N}_{12}$ ($=\text{Al}_3\text{Ga}_1\text{N}_4+\text{Al}_2\text{Ga}_1\text{N}_3+\text{Al}_5\text{Ga}_1\text{N}_6$)
- 11) $\text{Al}_{11}\text{Ga}_1\text{N}_{12}$ ($=\text{Al}_5\text{Ga}_1\text{N}_6+\text{AlN}$)

However, since these six AlGa₃N composition ratios of 6) to 11) are combinations of two AlGa₃N composition ratios among the first metastable AlGa₃N, GaN and AlN, located before and after them, the c-axis symmetry is likely to be disturbed, and the stability is lowered than the first metastable AlGa₃N. But these six AlGa₃N are more stable than AlGa₃N in a random asymmetric arrangement state because the symmetry of the atomic arrangement of Al and Ga in the A3 and B3 planes is the same as that in the first metastable AlGa₃N. Hereinafter, $\text{Al}_{x_2}\text{Ga}_{1-x_2}\text{N}$ of the AlN mole fraction x_2 ($x_2=1/12, 1/4, 1/2, 7/12, 3/4, 11/12$) corresponding to AlGa₃N composition ratios of above-mentioned 6) to 11) is referred to as the “second metastable AlGa₃N”, for convenience of explanation. As described above, the first and second metastable AlGa₃N have a stable structure due to the symmetry of the atomic arrangement of Al and Ga in the crystalline structure. Hereinafter, the first and second metastable AlGa₃N are collectively referred to as “metastable AlGa₃N”.

To grow AlGa₃N with maintaining constant crystal quality, it is required to perform crystal growth at a high temperature of 1000° C. or higher. However, it is assumed that Ga moves around at 1000° C. or higher even after atoms reach the sites of the crystal surface. On the other hand, since Al tends to adsorb to the surface unlike Ga, the movement after entering the site is strongly restricted though it is considered to move somewhat.

Therefore, even though $\text{Al}_1\text{Ga}_5\text{N}_6$ of above-mentioned 1), $\text{Al}_1\text{Ga}_{11}\text{N}_{12}$ of above-mentioned 6), and $\text{Al}_1\text{Ga}_3\text{N}_4$ of above-mentioned 7) are the metastable AlGa₃N, since AlN mole fractions are all less than or equal to 25% and Ga composition ratios are high, Ga movement is intense at a growth temperature around 1000° C., the symmetry of atomic

arrangement is disturbed, and the atomic arrangement of Al and Ga is close to random condition. As a result, it is considered that the stability described above is reduced compared with other metastable AlGa₃N.

Next, the “first Ga-rich region” will be described. In the nitride semiconductor ultraviolet light-emitting element and the manufacturing method of the nitride semiconductor ultraviolet light-emitting element with the above-described features, since each semiconductor layer in the n-type layer and the active layer is an epitaxially grown layer having a surface on which multi-step terraces parallel to the (0001) plane are formed, in the n-type layer, the easily mass-transferring Ga moves on the terrace region and concentrates on a boundary region between adjacent terraces, thereby forming a region having a lower AlN mole fraction than the terrace region. This boundary region extends obliquely upward with respect to the (0001) plane along with the epitaxial growth of n-type AlGa₃N layer of the n-type layer, so that stratiform regions with locally lower AlN mole fraction are formed with uniformly dispersed in the n-type layer. Here, if the amount of Ga mass-transferring is sufficiently large, the stratiform region may be the first Ga-rich region which includes the n-type AlGa₃N region of the metastable AlGa₃N, in which the AlGa₃N composition ratio is $\text{Al}_1\text{Ga}_1\text{N}_2$.

The metastable AlGa₃N, in which the AlGa₃N composition ratio is $\text{Al}_1\text{Ga}_1\text{N}_2$, is present within the first Ga-rich region, so that variation in amount of Ga supplied into the first Ga-rich region are absorbed in the metastable AlGa₃N. That is, in the first Ga-rich region, when the Ga supply amount increases, the metastable AlGa₃N increases, and when the Ga supply amount decreases, the metastable AlGa₃N decreases, and as a result, the variation of the AlN mole fraction in the first Ga-rich region is suppressed. Therefore, in the first Ga-rich region, the variation of the Ga supply amount due to the drift of the crystal growth apparatus or the like is absorbed, and the metastable AlGa₃N in which the AlGa₃N composition ratio is $\text{Al}_1\text{Ga}_1\text{N}_2$ (AlN mole fraction is 50%) is stably formed. That is, the variation of the AlN mole fraction in the first Ga-rich region is suppressed against the variation of the Ga supply amount.

However, as described above, since a state of the random asymmetric arrangement and a state of the regular symmetric arrangement can usually coexist in the crystal growth of AlGa₃N, a region of the metastable AlGa₃N in the state of the regular symmetric arrangement, in which the AlN mole fraction is 50%, is stably formed in the first Ga-rich region, and a region in which the AlN mole fraction fluctuates slightly (for example, about 0 to 3%) from 50% is also mixed therein. Therefore, the AlN mole fraction in the first Ga-rich region is concentrated and distributed near the AlN-mole fraction (50%) of the metastable AlGa₃N in which the AlGa₃N composition ratio is $\text{Al}_1\text{Ga}_1\text{N}_2$.

The first Ga-rich region, which is a stratiform region having a locally lower AlN mole fraction, is stably formed in the n-type layer, so that the carriers in the n-type layer are localized in the first Ga-rich region having a low band gap energy in the n-type layer, and a current can preferentially flow through the first Ga-rich region stably in the n-type layer, thereby suppressing variation in characteristics of the nitride semiconductor ultraviolet light-emitting element.

Next, the “well layer” and the “second Ga-rich region” will be described. Since each semiconductor layer in the n-type layer and the active layer is an epitaxially grown layer having a surface on which multi-step terraces parallel to the (0001) plane are formed, a boundary region between adjacent terraces of the multi-step terraces of the well layer is an

inclined region inclined with respect to the (0001) plane connecting between the adjacent terraces (see Non-Patent Documents 1 and 2 above). Incidentally, the inclined region consists of many steps (steps of one unit cell) and macrosteps (steps of multiple unit cells), and the (0001) surface shown stepwise on the inclined region is distinguished from the terrace surface of the multi-step terraces.

With growing of the side surfaces of the terrace edges toward lateral direction in step flow growth, a terrace on the upper surface of the well layer moves laterally relative to a terrace on the lower surface of the well layer, so that a film thickness of the well layer in the inclined region is getting thicker than that of the terrace region other than the inclined region. In addition, the second Ga-rich regions having locally lower AlN mole fraction are formed in the inclined regions in the well layer, as well as the first Ga-rich regions having locally lower AlN mole fraction are formed in the n-type layer. Here, when the amount of Ga mass-transferring is sufficiently large, an AlGa₂N₃ region of the metastable AlGa₂N₃ in which the AlGa₂N₃ composition ratio is Al₁Ga₂N₃ is formed in the second Ga-rich region.

The metastable AlGa₂N₃, in which the AlGa₂N₃ composition ratio is Al₁Ga₂N₃, is present in the second Ga-rich region, so that variation in Ga supplied into the second Ga-rich region are absorbed in the metastable AlGa₂N₃. That is, in the second Ga-rich region, when the Ga supply amount increases, the metastable AlGa₂N₃ increases, and when the Ga supply amount decreases, the metastable AlGa₂N₃ decreases, and as a result, the variation of the AlN mole fraction in the second Ga-rich region is suppressed. Accordingly, in the second Ga-rich region, the variation of the Ga supply amount due to the drift of the crystal growth apparatus or the like is absorbed, and the metastable AlGa₂N₃ in which the AlGa₂N₃ composition ratio is Al₁Ga₂N₃ (AlN mole fraction is 33.3% ($\frac{1}{3}$)) is stably formed. That is, the variation of the AlN mole fraction in the second Ga-rich region is suppressed against the variation of the Ga supply amount. In this specification, when an AlN mole fraction of $\frac{1}{3}$ is expressed as a percentage, it is approximately indicated as 33.3%.

However, as described above, since a state of the random asymmetric arrangement and a state of the regular symmetric arrangement can usually coexist in the crystal growth of AlGa₂N₃, a region of the metastable AlGa₂N₃ in the state of the regular symmetric arrangement, in which the AlN mole fraction is 33.3%, is stably formed in the second Ga-rich region, and a region in which the AlN mole fraction fluctuates slightly (for example, about 0 to 3%) from 33.3% is also mixed therein.

As described above, the band gap energy of the inclined region is smaller than that of the terrace region, and the localization of carriers is easy to occur as in the case of the first Ga-rich region of the n-type layer. For this reason, the light emission in the well layer is more remarkable in the inclined region than in the terrace region. In the above-mentioned Non-Patent Documents 1 and 2, the same content for the well layer of AlGa₂N₃-based semiconductor is reported. Incidentally, each terrace region of the well layer and the barrier layer is a region sandwiched between the terrace on the upper surface and the terrace on the lower surface of each layer in the c-axis direction. Therefore, portion other than each terrace region of the well layer and the barrier layer is the boundary region (the inclined region) of each layer.

Furthermore, the multi-step terraces formed by epitaxial growth of the active layer are formed successively from the multi-step terraces formed by epitaxial growth of the n-type layer. Therefore, the carriers (electrons) supplied to the well

layer along the current path in the first Ga-rich region are intensively supplied to the boundary region (the inclined region) between adjacent terraces where light emission is concentrated in the well layer.

Therefore, the n-type AlGa₂N₃ region, which is the metastable AlGa₂N₃ having an AlN mole fraction of 50%, is stably formed in the first Ga-rich region dominantly present in the stratiform region of the n-type layer, and furthermore, the AlGa₂N₃ region, which is the metastable AlGa₂N₃ having an AlN mole fraction of 33.3%, is stably formed in the second Ga-rich region in the inclined region of the well layer, whereby the carriers can be stably supplied to the inclined region of the well layer, and variation in characteristics such as the luminous efficiency of the nitride semiconductor ultraviolet light-emitting element can be suppressed.

Furthermore, since the n-type layer has an AlN mole fraction of 50% or more and ultraviolet rays emitted at the well layer of the active layer pass through the n-type layer, the element structure for extracting the ultraviolet light-emission from the n-type layer side can be adopted.

Furthermore, it is preferable that in the nitride semiconductor ultraviolet light-emitting element of the above-described feature, an AlN mole fraction of an n-type body region outside the stratiform rich region in the n-type layer is within a range of 54% to 66%.

Furthermore, in the manufacturing method of the nitride semiconductor ultraviolet light-emitting element of the above-described feature, it is preferable that, in the first step, a target AlN mole fraction of the n-type layer is set within a range of 54% to 66% and the n-type AlGa₂N₃ region in which an AlGa₂N₃ composition ratio is an integer ratio of Al₁Ga₁N₂ is grown in the first Ga-rich regions.

In these preferred implementations, the AlN mole fraction of the n-type body region outside the stratiform first Ga-rich region is within the range of 54% to 66% by absorbing the variation of the Ga supply amount due to the drift of the crystal growth apparatus or the like, so that the AlN mole fraction difference between the first Ga-rich region and the n-type body region is stably ensured to be 4% or more. Therefore, the carriers in the n-type layer are more stably localized within the first Ga-rich region having a smaller band gap energy than the n-type body region, and the current can flow stably in the first Ga-rich region preferentially in the n-type layer, thereby suppressing the characteristic variation of the nitride semiconductor ultraviolet light-emitting element.

Furthermore, since the upper limit of the AlN mole fraction in the n-type body region of the n-type layer and the upper limit of the target AlN mole fraction of the n-type layer are defined to be 66%, the metastable AlGa₂N₃ in which the AlGa₂N₃ composition ratio is Al₂Ga₁N₃ is not dominantly formed in the n-type layer. If the upper limit is 67% or more, the metastable AlGa₂N₃ of Al₂Ga₁N₃ is stably formed in the n-type body region, it is getting difficult to sufficiently supply Ga from the metastable AlGa₂N₃ to the first Ga-rich region to stably form the metastable AlGa₂N₃ of Al₁Ga₁N₂, an AlN mole fraction of the n-type AlGa₂N₃-based semiconductor formed in the first Ga-rich region will be randomly varied, and the desired effect cannot be expected.

Furthermore, in the nitride semiconductor ultraviolet light-emitting element of the above-described feature, it is preferable that an AlN mole fraction other than the boundary region parts in the well layer is within a range of 33.4% to 37%.

Furthermore, in the manufacturing method of the nitride semiconductor ultraviolet light-emitting element of the above-described feature, in the second step, it is preferable

to set a target AlN mole fraction of the well layer within a range of 33.4% to 37% and to grow AlGaIn region in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_2\text{In}_3$ in the second Ga-rich region.

According to these preferred implementations, the variation of the AlN mole fraction in the well layer is suppressed to 3.7% or less, and moreover, even when light emission peaks due to the compositional modulation generated from regions in which the AlN mole fraction is other than 33.3% overlap, it is possible to form quantum-wells exhibiting quasi-single-peak, so that the separation of the light emission peaks in the light emission spectrum is prevented.

Furthermore, in the nitride semiconductor ultraviolet light-emitting element of the above-described feature, it is preferable that the active layer has a multi-quantum-well structure including two or more well layers, and that a barrier layer composed of AlGaIn-based semiconductor is present between two of the well layers.

Furthermore, the manufacturing method of the nitride semiconductor ultraviolet light-emitting element of the above-described feature, in the second step, it is preferable to stack a well layer composed of an AlGaIn-based semiconductor and a barrier layer composed of an AlGaIn-based semiconductor alternately by epitaxial growth, and to form the active layer of a multi-quantum-well structure including two or more well layers, in which multi-step terraces parallel to the (0001) plane appear on each surface of the barrier layer and the well layer.

According to these preferred implementations, the active layer has a multi-quantum-well structure, and the luminous efficiency can be expected to be improved as compared with the luminous efficiency when only one well layer is used.

Furthermore, in the nitride semiconductor ultraviolet light-emitting element of the above-described feature, it is more preferable that the barrier layer is composed of an AlGaIn-based semiconductor, boundary region parts between adjacent terraces of the multi-step terraces of the barrier layer located at least on the most p-type layer side between two of the well layers has a third Ga-rich region with a locally lower AlN mole fraction within the same barrier layer.

Furthermore, in the manufacturing method of the nitride semiconductor ultraviolet light-emitting element of the above-described feature, when forming the barrier layer composed of an AlGaIn-based semiconductor in the second step, it is more preferable to form a third Ga-rich region having a locally lower AlN mole fraction within the same barrier layer in boundary region parts between the terraces of the barrier layer located on at least the most p-type layer side between two of the well layers.

According to these preferred implementations, the carrier localization can occur in the third Ga-rich region of the barrier layer as well as in the first Ga-rich regions of the n-type layer and the second Ga-rich region of the well layer. Therefore, when supplying the carriers (electrons) from the n-type layer to the second Ga-rich region of the boundary region (the inclined region) between adjacent terraces in which light emission is concentrated in the well layer, it can be done efficiently through the first Ga-rich regions of the n-type layer and the third Ga-rich regions of the barrier layer.

Here, in the multi-quantum-well structure having two or more well layers, since the emission intensity is large in the well layer at the most p-type layer side, and the third Ga-rich region is formed in the barrier layer at the n-type layer side

of the above-mentioned well layer, it can be done more efficiently to supply the carriers to the well layer as described above.

Furthermore, in the nitride semiconductor ultraviolet light-emitting element of the above preferred implementation, it is preferable that an AlGaIn region in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_1\text{In}_2$, $\text{Al}_2\text{Ga}_1\text{In}_3$, $\text{Al}_3\text{Ga}_1\text{In}_4$ or $\text{Al}_5\text{Ga}_1\text{In}_6$ exists in the third Ga-rich region of the barrier layer.

Furthermore, in the manufacturing method of the nitride semiconductor ultraviolet light-emitting element of the above preferred implementation, it is preferable in the second step,

- 1) to grow an AlGaIn region in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_1\text{In}_2$ in the third Ga-rich region by setting a target AlN mole fraction of the barrier layer within a range of 51% to 66%, or
- 2) to grow an AlGaIn region in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_2\text{Ga}_1\text{In}_3$ in the third Ga-rich region by setting a target AlN mole fraction of the barrier layer within a range of 68% and 74%, or
- 3) to grow an AlGaIn region in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_3\text{Ga}_1\text{In}_4$ in the third Ga-rich region by setting a target AlN mole fraction of the barrier layer within a range of 76% to 82%, or
- 4) to grow an AlGaIn region in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_5\text{Ga}_1\text{In}_6$ in the third Ga-rich region by setting a target AlN mole fraction of the barrier layer within a range of 85% to 90%.

According to these preferred implementations, since the metastable AlGaIn is present in the third Ga-rich region of the barrier layer, the variation of the AlN mole fraction of the third Ga-rich region is suppressed as well as the first Ga-rich region of the n-type layer and the second Ga-rich region of the well layer, and the region of the metastable AlGaIn is stably formed in the third Ga-rich region. Therefore, the effects achieved by the third Ga-rich region of the barrier layer are more stably realized.

Furthermore, it is preferable that the nitride semiconductor ultraviolet light-emitting element of the above features further comprises an underlying part containing a sapphire substrate, the sapphire substrate has a main surface inclined by a predetermined angle with respect to the (0001) plane, the light-emitting element structure part is formed above the main surface, each semiconductor layer at least from the main surface of the sapphire substrate to the surface of the active layer is an epitaxially grown layer having a surface on which multi-step terraces parallel to the (0001) plane are formed.

According to the preferred implementation, a miscut sapphire substrate can be used to perform epitaxial growth such that multi-step terraces appear on the surface of each layer from the main surface of the sapphire substrate to the surface of the active layer, thereby realizing the nitride semiconductor ultraviolet light-emitting element of the above-described feature.

Effect of the Invention

According to the nitride semiconductor ultraviolet light-emitting element and the manufacturing method of the nitride semiconductor ultraviolet light-emitting element of the above-described features, it is possible to stably provide a nitride semiconductor ultraviolet light-emitting element

13

having a desired emission characteristic with suppressed characteristic variation caused by the drift of the crystal growth apparatus or the like.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram schematically illustrating the wurtzite crystal structure of AlGaIn.

FIG. 2 is a plan view showing the positional relationship between each site of A plane and each site of B plane as viewed from the c-axis direction of the wurtzite crystal structure shown in FIG. 1.

FIG. 3 is a diagram schematically showing the arrangement of Al and Ga on A3 plane and B3 plane in each of five combinations of AlGaIn composition ratios represented by integer ratios.

FIG. 4 is a fragmentary cross-sectional view schematically showing an exemplary configuration of a nitride semiconductor ultraviolet light-emitting element according to an embodiment of the present invention.

FIG. 5 is a fragmentary cross-sectional view schematically showing an exemplary laminated structure of an active layer of the nitride semiconductor ultraviolet light-emitting element shown in FIG. 1.

FIG. 6 is a graph showing the relationship between the emission wavelength of the quantum-well structure consisting of AlGaIn well layer and AlGaIn barrier layer, the film thickness of the well layer, and the AlN mole fraction of the barrier layer.

FIG. 7 is a plan view schematically showing an exemplary structure when the nitride semiconductor ultraviolet light-emitting element shown in FIG. 4 is viewed from the upper side of FIG. 4.

FIG. 8 is a HAADF-STEM image showing the cross-sectional construction in the n-type cladding layer.

FIG. 9 is a diagram illustrating six measurement regions A to F, in which line-analysis of the cross-sectional TEM-EDX in the n-type cladding layer is performed, in the HAADF-STEM image shown in FIG. 8.

FIG. 10A is a diagram showing measurement results of AlN mole fraction and GaN mole fraction by the line-analysis of the cross-sectional TEM-EDX in the n-type cladding layer in the measurement region A shown in FIG. 9.

FIG. 10B is a diagram showing measurement results of AlN mole fraction and GaN mole fraction by the line-analysis of the cross-sectional TEM-EDX in the n-type cladding layer in the measurement region B shown in FIG. 9.

FIG. 10C is a diagram showing measurement results of AlN mole fraction and GaN mole fraction by the line-analysis of the cross-sectional TEM-EDX in the n-type cladding layer in the measurement region C shown in FIG. 9.

FIG. 10D is a diagram showing measurement results of AlN mole fraction and GaN mole fraction by the line-analysis of the cross-sectional TEM-EDX in the n-type cladding layer in the measurement region D shown in FIG. 9.

FIG. 10E is a diagram showing measurement results of AlN mole fraction and GaN mole fraction by the line-analysis of the cross-sectional TEM-EDX in the n-type cladding layer in the measurement region E shown in FIG. 9.

FIG. 10F is a diagram showing measurement results of AlN mole fraction and GaN mole fraction by the line-

14

analysis of the cross-sectional TEM-EDX in the n-type cladding layer in the measurement region F shown in FIG. 9.

FIG. 11 is a SEM image showing the measurement regions of the AlN mole fraction by the CL method in the n-type cladding layer.

FIG. 12 is a diagram showing the first and second CL spectra calculated from ten CL spectra measured in each measurement region shown in FIG. 11.

FIG. 13 is a diagram showing CL spectra of the inclined region and the terrace region of the well layer having the multi-quantum-well structure shown in FIG. 5.

FIG. 14 is a fragmentary cross-sectional view schematically showing an example of an element structure of a general ultraviolet light-emitting diode.

DESCRIPTION OF EMBODIMENT

A nitride semiconductor ultraviolet light-emitting element (hereinafter, simply referred to as a "light-emitting element") according to an embodiment of the present invention will be described with reference to the drawings. In the drawings as pattern diagram used in the following description, the dimensional ratios of each part are not necessarily the same as those of the actual elements because the essential part is emphasized to schematically show the invention for ease of understanding of the description. Hereinafter, in the present embodiment, description will be made on the assumption that the light-emitting element is a light-emitting diode.

First Embodiment

<Element Structure of Light-Emitting Element>

As shown in FIG. 4, the light-emitting element 1 of the present embodiment includes an underlying part 10 including a sapphire substrate 11, and a light-emitting element structure part 20 including a plurality of AlGaIn-based semiconductor layers 21-25, a p-electrode 26, and an n-electrode 27. The nitride semiconductor light-emitting element 1 is mounted (flip-chip mounted) on a mounting base (a submount or the like) with facing a side of the light-emitting element structure part 20 (an upper side in FIG. 4) toward the mounting base, and light is extracted from a side of the underlying part 10 (a lower side in FIG. 4). In this specification, for convenience of explanation, a direction perpendicular to the main surface 11a of the sapphire substrate 11 (or the upper surface of the underlying part 10 and the respective AlGaIn-based semiconductor layers 21 to 25) is referred to as "up and down direction" (or "vertical direction"), and a direction from the underlying part 10 to the light-emitting element structure part 20 is set to an upward direction and an opposite direction thereof is defined as a downward direction. A plane parallel to the vertical direction is referred to as a "first plane". A plane parallel to the main surface 11a of the sapphire substrate 11 (or the upper surface of the underlying part 10 and the respective AlGaIn-based semiconductor layers 21 to 25) is referred to as a "second plane," and a direction parallel to the second plane is referred to as a "lateral direction."

The underlying part 10 is configured with the sapphire substrate 11 and an AlN layer 12 formed directly on the main surface 11a of the sapphire substrate 11. The sapphire substrate 11 is a slightly inclined substrate, in which the main surface 11a is inclined at an angle (miscut angle) within a certain range (e.g., from 0 degree to about 6

15

degrees) with respect to the (0001) plane and the multi-step terraces appear on the main surface **11a**.

The AlN layer **12** is composed of AlN crystal epitaxially grown on the main surface of the sapphire substrate **11**, the AlN crystal has an epitaxial crystal orientation relationship with respect to the main surface **11a** of the sapphire substrate **11**. Specifically, for example, the AlN crystal is grown so that the C-axis direction of the sapphire substrate **11** (<0001> direction) and the C-axis direction of the AlN crystal is aligned. The AlN crystal constituting AlN layer **12** may be an AlN-based semiconductor layer which contains a trace amount of Ga or another impurity. In the present embodiment, the film thickness of the AlN layer **12** is assumed to be about 2 μm to 3 μm . The structure of the underlying part **10** and a substrate to be used are not limited to those described above. For example, an AlGaIn-based semiconductor layer having an AlN mole fraction greater than or equal to the AlN mole fraction of the AlGaIn-based semiconductor layer **21** may be provided between the AlN layer **12** and the AlGaIn-based semiconductor layer **21**.

The AlGaIn-based semiconductor layers **21-25** of the light-emitting element structure part **20** comprises a structure having an n-type cladding layer **21** (n-type layer), an active layer **22**, an electron blocking layer **23** (p-type layer), a p-type cladding layer **24** (p-type layer), and a p-type contact layer **25** (p-type layer) stacked in order from the underlying part **10** side by epitaxially growing them in order.

In the present embodiment, each semiconductor layer within the AlN layer **12** of the underlying part **10**, the n-type cladding layer **21** and the active layer **22** of the light-emitting element structure part **20** which are epitaxially grown in order from the main surface **11a** of the sapphire substrate **11** has a surface on which multi-step terraces parallel to the (0001) plane originating from the main surface **11a** of the sapphire substrate **11** are formed. Since the p-type layers of the electron blocking layer **23**, the p-type cladding layer **24**, and the p-type contact layer **25** are formed on the active layer **22** by epitaxial growth, the same multi-step terraces can be formed, but the p-type layers do not necessarily need to have surfaces on which the same multi-step terraces are formed.

As shown in FIG. 4, in the light-emitting element structure part **20**, the active layer **22**, the electron blocking layer **23**, the p-type cladding layer **24**, and the p-type contact layer **25** are formed on the first region R1 of the upper surface of the n-type cladding layer **21** as a result of removing portions of them stacked on the second region R2 of the upper surface of the n-type cladding layer **21** by etching or the like. The upper surface of the n-type cladding layer **21** is exposed in the second region R2 except for the first region R1. The upper surface of the n-type cladding layer **21** may differ in height between the first region R1 and the second region R2, as schematically shown in FIG. 4, where the upper surface of the n-type cladding layer **21** is individually defined in the first region R1 and the second region R2.

The n-type cladding layer **21** is composed of the n-type AlGaIn-based semiconductor, and stratiform regions, in which AlN mole fraction is locally lower within the n-type cladding layer **21**, are present uniformly dispersed in the n-type cladding layer **21**. As described above, the first Ga-rich region **21a**, which includes the n-type AlGaIn region in which the AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_1\text{N}_2$ (i.e., the n-type metastable AlGaIn with an AlN mole fraction of 50%), is dominantly present in the stratiform regions. FIG. 4 schematically shows the stratiform region which is entirely the first Ga-rich region **21a**, as examples in which the first Ga-rich region **21a** is dominantly

16

present in the stratiform region. The region other than the stratiform region in the n-type cladding layer **21** is referred to as an n-type body region **21b**.

In the present embodiment, the AlN mole fraction of the n-type body region **21b** is adjusted within a range of 54% to 66%. As the film thickness of the n-type cladding layer **21** is assumed to be about 1 μm to 2 μm , as with the film thickness adopted in the common nitride semiconductor ultraviolet light-emitting element, but the film thickness may be about 2 μm to 4 μm . In the following, for brevity purposes, the n-type AlGaIn region of the metastable AlGaIn, in which the AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_1\text{N}_2$, present in the first Ga-rich region **21a** is referred to as a "metastable n-type region" for convenience. A region other than the metastable n-type region present in the first Ga-rich region **21a**, in which the AlN mole fraction slightly fluctuates with respect to 50% ($1/2$), is referred to as a "metastable neighbor n-type region". Here, the metastable n-type region does not necessarily need to be continuously stratified in the plurality of stratiform first Ga-rich regions **21a** and may intermittently exist separated by the metastable neighbor n-type region.

The active layer **22** has a multi-quantum-well structure in which two or more well layers **220** composed of an AlGaIn-based semiconductor, and one or more barrier layers **221** composed of an AlGaIn-based semiconductor or an AlN-based semiconductor are stacked alternately. The barrier layer **221** is not necessarily provided between the lowermost well layer **220** and the n-type cladding layer **21**. In addition, the barrier layer **221** or a layer of AlGaIn or AlN having a thinner thickness and a higher AlN mole fraction than the barrier layer **221** may be provided between the uppermost well layer **220** and the electron blocking layer **23**.

FIG. 5 schematically shows an exemplary laminated structure (multi-quantum-well structure) of the well layer **220** and the barrier layer **221** in the active layer **22**. FIG. 5 illustrates an example for three well layers **220**. The structure in which the terrace T in the well layer **220** and the barrier layer **221** shown in FIG. 5 grows in multi-steps is a known structure as disclosed in Non-Patent Documents 1 and 2 described above. The boundary region BA between the adjacent terraces T has an inclined region inclined with respect to the (0001) plane as described above. In the present embodiment, the tread (distance between adjacent boundary regions BA) of one terrace T is assumed to be several tens nm to several hundreds nm.

As schematically shown in FIG. 5, the second Ga-rich region **220a** having a locally lower AlN mole fraction in the well layer **220** is formed in the boundary region (the inclined region) BA between the adjacent terraces T of the multi-step terraces T in the well layer **220**. A region other than the second Ga-rich region **220a** in the well layer **220** is referred to as a well body region **220b** for convenience.

In the present embodiment, the metastable AlGaIn in which the AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_2\text{N}_3$, that is, AlGaIn having an AlN mole fraction of 33.3% ($1/3$) is present in the second Ga-rich region **220a**. Also, the AlN mole fraction of the well body region **220b** is adjusted to be within 33.4% to 37%. The film thickness of the well layer **220**, including the terrace region TA and the inclined region BA, is adjusted within a range of 2 unit cells to 7 unit cells, for example.

In the following, for brevity purposes, a metastable AlGaIn, in which the AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_2\text{N}_3$, present in the second Ga-rich region **220a** is referred to as a "metastable well region" for convenience. A region other than the metastable well region present in the

second Ga-rich region **220a**, in which the AlN mole fraction slightly fluctuates with respect to 33.3% ($\frac{1}{3}$), is referred to as a “metastable neighbor well region”. Here, in the second Ga-rich region **220a** formed in the inclined region BA existing along an edge line of the terrace T in planar view, the metastable well region does not need to exist continuously along the edge line and may exist intermittently separated by the metastable neighbor n-type region.

As described above, the barrier layer **221** is composed of an AlGaIn-based semiconductor and has a surface on which multi-step terraces T parallel to the (0001) plane are formed as with the n-type cladding layer **21** and the well layer **220**. Here, the AlN mole fraction of the entire barrier layer **221** is assumed to be in a range of 50% to 100%, for example. The barrier layer **221** may be composed of an AlGaIn-based semiconductor in which the AlN mole fraction is not 100%, although the barrier layer **221** may be composed of an AlN-based semiconductor in which the AlN mole fraction is 100%. Therefore, as schematically shown in FIG. 5, when the barrier layer **221** is composed of an AlGaIn-based semiconductor in which AlN mole fraction is not 100%, the third Ga-rich region **221a** having a locally lower AlN mole fraction in the barrier layer **221** can be formed in the boundary region (the inclined region) BA between the adjacent terraces T of the barrier layer **221**, similarly to the n-type cladding layer **21** and the well layer **220**. A region other than the third Ga-rich region **221a** of the terrace region in the barrier layer **221** is referred to as a barrier body region **221b** for convenience. The barrier body region **221b** mainly exists in the terrace region TA in the barrier layer **221**. If the AlN mole fraction of the entire barrier layer **221** including the third Ga-rich region **221a** is, as an example, in the range of 50% to 90%, which is a part of the above-mentioned range of 50% to 100%, it is preferable that the AlN mole fraction difference between the third Ga-rich region **221a** and the barrier body region **221b** is 4% to 5% or more in order to sufficiently secure the effect of the carrier localization in the third Ga-rich region **221a**, but the effect of the carrier localization can be expected even if the AlN mole fraction difference is about 1%. Therefore, in this embodiment, the AlN mole fraction of the barrier body region **221b** is set within a range of 51% to 90%. Furthermore, it is preferable that the thickness of the barrier layer **221** is adjusted within a range of, for example, 6 nm to 8 nm, including the terrace region TA and the inclined region BA.

FIG. 6 is a graphical representation of the simulated emission wavelengths (corresponding to the peak emission wavelengths) obtained by changing the film thickness of the well layer within a range of 4 ML (monolayer) to 14 ML (2 unit cells to 7 unit cells) for the quantum-well structure model in which the well layer **220** and the barrier layer **221** are composed of AlGaIn. As a condition of the above simulation, the AlN mole fraction of the second Ga-rich region **220a** of the well layer **220** was set to 33.3% ($\frac{1}{3}$) which is an AlN mole fraction of the metastable well region, and three AlN mole fractions of the third Ga-rich region **221a** of the barrier layer **221** were set to 66.7%, 75%, and 83.3%. In the simulated result shown in FIG. 6, it is assumed that the ultraviolet light emission in the well layer **220** is remarkably generated in the boundary region (the inclined region) BA. For this reason, it is essential that the film thickness condition of the well layer **220** is satisfied in the inclined region BA.

From FIG. 6, the quantum confinement effect in the well layer **220** is increased and the emission wavelength is shortened as the film thickness of the well layer **220** is decreased within the range of 4 ML to 14 ML. Also, the

degree of change in the emission wavelength with respect to the change in the film thickness of the well layer **220** increases as the AlN mole fraction of the barrier layer **221** is increased. It can also be seen that within the above ranges of the film thickness of well layer **220** and the AlN mole fraction of the barrier layer **221**, the emission wavelength varies in the range of approximately 261 nm to 328 nm. Furthermore, the barrier layer **221** can be made of AlN to further extend the emission wavelength. Emission wavelengths of 277 nm and 315 nm indicated by dashed-dotted lines in FIG. 6 are control ranges (lower limit and upper limit) of the emission wavelength assumed in the light-emitting element **1** of the present embodiment. Because of the dominant presence of the metastable AlGaIn in which the AlGaIn composition ratio is $\text{Al}_1\text{Ga}_2\text{N}_3$ in the second Ga-rich region **220a** of the well layer **220**, the emission wavelength can be controlled within a range of 277 nm to 315 nm by setting the film thickness of the well layer **220** within a range of 5 ML to 14 ML in accordance with the AlN mole fraction of the barrier layer **221**, as an example.

The electron blocking layer **23** is composed of a p-type AlGaIn-based semiconductor. The p-type cladding layer **24** is composed of a p-type AlGaIn-based semiconductor. The p-type contact layer **25** is composed of a p-type AlGaIn-based semiconductor or p-type GaN based semiconductor. The p-type contact layer **25** is typically composed of GaN. The thicknesses of the respective layers such as the active layer **22**, the electron blocking layer **23**, the p-type cladding layer **24**, and the p-type contact layer **25** are appropriately determined in accordance with the emission wavelength characteristics and the electric characteristics of the light-emitting element **1**. The p-type cladding layer **24** may be omitted in order to reduce the parasitic resistance of the p-type layers.

The p-electrode **26** is made of, for example, a multilayer metal film such as Ni/Au, and is formed on the upper surface of the p-type contact layer **25**. The n-electrode **27** is made of, for example, a multilayer metal film such as Ti/Al/Ti/Au and is formed on a part of the exposed surface of the n-type cladding layer **21** in second region R2. The p-electrode **26** and the n-electrode **27** are not limited to the multilayer metal film described above, and the electrode structure such as the metal constituting each electrode, the number of layers, and the stacking order of layers may be changed as appropriate. FIG. 7 shows an example of shapes of the p-electrode **26** and the n-electrode **27** viewed from the upper side of the light-emitting elements **1**. In FIG. 7, a line BL existing between the p-electrode **26** and the n-electrode **27** represents a boundary line between the first region R1 and the second region R2 and coincides with the outer peripheral side wall surfaces of the active layer **22**, the electron blocking layer **23**, the p-type cladding layer **24**, and the p-type contact layer **25**.

Although in the present embodiment, as shown in FIG. 7, a comb-like shape is employed as an example of the planarly-viewed shapes of the first region R1 and the p-electrode **26**, the planarly-viewed shapes and arrangements of the first region R1 and the p-electrode **26** are not limited to the illustration shown in FIG. 7.

When a forward bias is applied between the p-electrode **26** and the n-electrode **27**, holes are supplied from the p-electrode **26** toward the active layer **22**, electrons are supplied from the n-electrode **27** toward the active layer **22**, and the supplied holes and electrons respectively reach the active layer **22** and recombine to emit light. This also causes a forward current to flow between the p-electrode **26** and the n-electrode **27**.

As to the first Ga-rich region **21a** of the n-type cladding layer **21**, as shown schematically by a double line in FIG. 4, a plurality of layers is vertically separated from each other. Furthermore, in one first plane (for example, the cross section shown in FIG. 4) which is parallel to the vertical direction, at least a part of the extending direction of the first Ga-rich region **21a** is inclined with respect to the lateral direction (the extending direction of an intersection line of the first plane and the second plane). In the first plane shown in FIG. 4, each layer of the first Ga-rich regions **21a** is schematically illustrated by a parallel line (double line), the inclination angle θ formed between the extending direction and the lateral direction is not necessarily the same between the first Ga-rich regions **21a**, and may vary depending on the position even within the same first Ga-rich region **21a**, so that the first Ga-rich region **21a** on the first plane do not necessarily extend linearly. Furthermore, the inclination angle θ is also changed by the orientation of the first plane. Therefore, a part of the first Ga-rich region **21a** may intersect with or diverge from another first Ga-rich region **21a** on the first plane. It is clearly indicated in HAADF-STEM image shown in FIG. 8 that the inclination angle formed between the extending direction and the lateral direction of the first Ga-rich region **21a** varies depending on the position, and that the first Ga-rich region **21a** is uniformly dispersed in the n-type cladding layer **21**.

The first Ga-rich region **21a** is shown as one line (double line) on the first plane in FIG. 4, but also extends in a direction perpendicular to the first plane in a direction parallel or inclined to the second plane and has a two-dimensional extension. Accordingly, the plurality of first Ga-rich regions **21a** exist in stripes on the plurality of second planes in the n-type cladding layer **21**.

The first Ga-rich region **21a** is a stratiform region with locally lower AlN mole fraction in the n-type cladding layer **21**, as described above. That is, the AlN mole fraction of the first Ga-rich region **21a** is relatively low with respect to the AlN mole fraction of the n-type body region **21b**. In addition, when the AlN mole fractions of both regions are asymptotically consecutive in the vicinity of the boundary between the first Ga-rich region **21a** and the n-type body region **21b**, the boundary between both regions cannot be clearly defined.

Therefore, in such cases, a portion in which the AlN mole fraction is lower than a reference value can be relatively defined as the first Ga-rich region **21a**, assuming that the reference value is the average AlN mole fraction of the entire n-type cladding layer **21**, for example, the AlN mole fraction serving as a basis for the growth condition of the n-type cladding layer **21** (supply amount and flow rate of the source gases and the carrier gas used in the metalorganic vapor phase epitaxy method), which will be described later. In addition to the above-mentioned defining methods, for example, based on a HAADF-STEM image to be described later, a portion having a large brightness change may be defined as the boundary between both layers. However, in the present invention, the definition of the boundary between both layers is not significant, and it is sufficient if the presence of the first Ga-rich region **21a** itself can be grasped.

Indeed, since the first Ga-rich region **21a** is formed with the mass-transfer of Ga from the n-type body region **21b**, the average AlN mole fraction in the first Ga-rich region **21a** varies depending on the amount of Ga supplied from the n-type body region **21b**, and the AlN mole fraction is not necessarily uniform even in the first Ga-rich region **21a**. However, in the present embodiment, since the metastable n-type region is stably formed in the first Ga-rich region **21a**,

even if there is a small variation in the above Ga supply amount, the variation is absorbed by the metastable n-type region, and the variation of the AlN mole fraction in the first Ga-rich region **21a** is suppressed. Therefore, the minimum value of the AlN mole fraction in each of the first Ga-rich regions **21a** is 50%, which is the AlN mole fraction of the metastable n-type regions, or a value in the vicinity thereof. However, as described above, the metastable neighbor n-type region also exists together with the metastable n-type region in the first Ga-rich region **21a**. Since the metastable neighbor n-type region is also formed with the mass-transfer of Ga from the n-type body region **21b**, usually, the AlN mole fraction of the metastable neighbor n-type region is higher than the AlN mole fraction of the metastable n-type region, and the average AlN mole fraction in the first Ga-rich region **21a** is slightly higher than the AlN mole fraction of the metastable n-type region.

On the other hand, in the n-type body region **21b**, Ga is supplied to the first Ga-rich region **21a**, so that the AlN mole fraction is relatively high at a portion in the n-type body region **21b** where Ga has been mass-transferred. Furthermore, the mass-transfer of Ga that does not lead to the formation of the first Ga-rich region **21a** may also occur in the n-type body region **21b**, so that the AlN mole fraction also varies to some extent in the n-type body region **21b**. However, as described above, since the carriers in the n-type cladding layer **21** is localized in the first Ga-rich region **21a** with smaller band gap energy than the n-type body region **21b**, and in the n-type cladding layer **21**, the current flows stably preferentially in the first Ga-rich region **21a**, even if the AlN mole fraction in the n-type body region **21b** varies slightly, the characteristic variation of the light-emitting element **1** is suppressed by the first Ga-rich region **21a**.

Here, the above explanation of the first Ga-rich region **21a** also applies to the second Ga-rich region **220a** as it is. In other words, in the present embodiment, since the metastable well region is stably formed in the second Ga-rich region **220a**, even if there is a small variation in the Ga supply amount, the variation is absorbed by the metastable well region, and the average AlN mole fraction of the second Ga-rich region **220a** is 33.3%, which is the AlN mole fraction of the metastable well region, or the vicinity thereof. However, as described above, the metastable neighbor well region also exists together with the metastable well region in the second Ga-rich region **220a**. Since the metastable neighbor well region is also formed with the mass-transfer of Ga from the well body region **220b**, usually, the AlN mole fraction of the metastable neighbor well region is higher than the AlN mole fraction of the metastable well region, and the average AlN mole fraction in the second Ga-rich region **220a** is slightly higher than the AlN mole fraction of the metastable well region.

On the other hand, when the well body region **220b** supplies Ga to the second Ga-rich region **220a**, the AlN mole fraction of Ga in the well body region **220b** is relatively high at a position after Ga is mass-transferred. Furthermore, the AlN mole fraction also varies to some extent in the well body region **220b**, since the mass-transfer of Ga may occur to a degree that does not lead to the second Ga-rich region **220a** being formed in the well body region **220b**. However, as described above, even if the AlN mole fraction in the well body region **220b** varies slightly, the characteristic variation of the light-emitting element **1** is suppressed by the second Ga-rich region **220a** because the carriers in the well layer **220** are localized in the second Ga-rich region **220a** having

21

a smaller bandgap energy than the well body region **220b** and the current flows through the second Ga-rich region **220a** preferentially

<Method for Manufacturing Light-Emitting Element>

Next, an example of a manufacturing method of the light-emitting element **1** illustrated in FIG. **4** will be described.

First, by a well-known epitaxial growth method such as metalorganic vapor phase epitaxy (MOVPE) method, the AlN layer **12** contained in underlying part **10** and the nitride semiconductor layers **21** to **25** contained in the light-emitting element structure part **20** are epitaxially grown on the sapphire substrate **11** sequentially and laminated. At this time, for example, Si is doped into the n-type cladding layer **21** as a donor impurity, and Mg is doped into the electron blocking layer **23**, the p-type cladding layer **24**, and the p-type contact layer **25** as an acceptor impurity.

In the present embodiment, in order to make multi-step terraces parallel to the (0001) plane on at least the AlN layers **12**, the n-type cladding layer **21**, and the active layer **22** (the well layer **220**, the barrier layer **221**), a slightly inclined substrate is used as the sapphire substrate **11**, in which the main surface **11a** is inclined at an angle (miscut angle) within a certain range (for example, from 0 degree to about 6 degrees) with respect to the (0001) plane, and the multi-step terraces appear on the main surface **11a**.

As a condition of such epitaxial growth, in addition to the use of the slightly inclined (0001) sapphire substrate **11**, for example, growth rate in which the multi-step terraces easily appear (specifically, for example, the growth rate achieved by appropriately setting various conditions such as growth temperature, supply amount and flow rate of the source gas and the carrier gas) and the like are included. Note that these various conditions may differ depending on the type and structure of the film forming apparatus, and therefore, it is good to actually make some samples in the film forming apparatus in order to specify these conditions.

As growth conditions of the n-type cladding layer **21**, a growth temperature, a growth pressure, and a donor impurity concentration are selected so that the growth start points of the first Ga-rich region **21a** are formed on the step portions (boundary region) between the multi-step terraces formed on the upper surface of the AlN layer **12** by the mass transfer of Ga immediately after the growth start, and the first Ga-rich region **21a** can grow obliquely upward by the segregation due to the mass transfer of Ga in accordance with the epitaxial growth of the n-type cladding layer **21** (the base layer **21a**).

Specifically, the growth temperature is preferably 1050° C. or higher at which the mass transfer of Ga easily occurs, and the growth temperature is preferably 1150° C. or lower at which a good n-type AlGaIn can be prepared. Furthermore, when the growth temperature exceeds 1170° C., the mass-transfer of Ga becomes excessive, and the AlN mole fraction tends to vary randomly even in the first metastable AlGaIn. Therefore, such a high growth temperature exceeding 1170° C. is not preferable since it is difficult to form stably the metastable AlGaIn in which the AlN mole fraction is 50%. The growth pressure of 75 Torr or less is preferable as the growth condition of a good AlGaIn, and the growth pressure of 10 Torr or more is practical as the control limit of the film forming apparatus. The donor impurity density is preferably about 1×10^{18} to 5×10^{18} cm⁻³. The above-described growth temperature, growth pressure, and the like are examples, and the optimum conditions may be appropriately specified according to the film forming apparatus to be used.

22

The supply amount and the flow rate of the source gases (trimethylaluminum (TMA) gas, trimethylgallium (TMG) gas, and ammonia gas) and the carrier gas used in the metalorganic vapor phase epitaxy method are set according to the average AlN mole fraction X_a of the entire n-type cladding layer **21** as a target value. Here, assuming that the average AlN mole fraction of the n-type body region **21b** is X_b (=54% to 66%), and the average AlN mole fraction of the first Ga-rich region **21a**, in which the metastable n-type region having the AlN mole fraction of 50% and the metastable neighbor n-type region having the AlN mole fraction slightly higher than 50%, is X_c (>50%), and considering the mass-transfer of Ga from the n-type body region **21b** to the first Ga-rich region **21a**, $X_b > X_a > X_c$. However, since the volume ratio of the first Ga-rich region **21a** to the entire n-type cladding layer **21** is small, it can be approximately set as $X_a = X_b$.

In the first Ga-rich region **21a**, the metastable n-type region having the AlN mole fraction of 50% is stably present, and since the target value X_a of the AlN mole fraction of the n-type cladding layer **21** is 54% to 66%, the difference between the AlN mole fraction of 50% in the metastable n-type region and the average AlN mole fraction X_b of the n-type body region **21b** ($X_b - 50\%$) is stably 4% or more, and the carriers in the n-type layer are localized in the first Ga-rich region **21a** having a smaller bandgap energy than the n-type body region **21b**. Furthermore, since the upper limit of the target value X_a is 66%, the metastable AlGaIn in which the AlGaIn composition ratio is $Al_2Ga_1N_3$ is not dominantly formed in the n-type body region **21b**. If the upper limit of the target value X_a is 67% or more, the metastable AlGaIn of $Al_2Ga_1N_3$ is stably formed in the n-type body region **21b**, and it is difficult to fully supply Ga from the metastable AlGaIn of $Al_2Ga_1N_3$ in order to stably form the metastable AlGaIn (the metastable n-type region) of $Al_1Ga_1N_2$ in the first Ga-rich region. Therefore, the metastable n-type region having the AlN mole fraction of 50% can be stably formed in the first Ga-rich region **21a** by setting the upper limit of the target value X_a to 66%.

Note that the donor impurity concentration does not necessarily have to be uniformly controlled in the vertical direction with respect to the film thickness of the n-type cladding layer **21**. For example, there may be a low impurity concentration layer in which the impurity concentration of a predetermined thin film thickness portion in the n-type cladding layer **21** is lower than the above-mentioned set concentration, and is controlled to be, for example, less than 1×10^{18} cm⁻³, more preferably, 1×10^{17} cm⁻³ or less. As the film thickness of the low impurity concentration layer, it is preferably about larger than 0 nm and 200 nm or less, more preferably about 10 nm or more and 100 nm or less, and still more preferably about 20 nm or more and 50 nm or less. Furthermore, the donor impurity concentration of the low impurity concentration layer may be lower than the set concentration, and an undoped layer (0 cm⁻³) may be partially included. Furthermore, it is preferable that a part of or all the low-impurity-concentration layer is present in the upper layer region having a depth of 100 nm or less at lower side from the upper surface of the n-type cladding layer **21**.

As described above, when the n-type cladding layer **21** having the first Ga-rich region **21a** and the n-type body region **21b** is formed, the active layer **22** (the well layer **220**, the barrier layer **221**), the electron blocking layer **23**, the p-type cladding layer **24**, the p-type contact layer **25**, and the like are subsequently formed on the entire upper surface of

the n-type cladding layer **21** by a well-known epitaxial growth method such as a metalorganic vapor phase epitaxy (MOVPE) method.

In forming the active layer **22**, under the growth conditions in which the above-described multi-step terraces easily appear in the same manner as in the n-type cladding layer **21**, the well layer **220** is grown using the AlN mole fraction (33.4% to 37%) set for the well body region **220b** as a target value, and the barrier layer **221** is grown using the AlN mole fraction (51% to 90% or 100%) set for the barrier body region **221b** as a target value.

Next, by a well-known etching method such as reactive ion etching, the second region **R2** of the nitride semiconductor layers **21** to **25** stacked in the above manner is selectively etched until the upper surface of the n-type cladding layer **21** is exposed. As a result, the second region **R2** part of the upper surface of the n-type cladding layer **21** is exposed. Then, the p-electrode **26** is formed on the p-type contact layer **25** in the unetched first region **R1** and the n-electrode **27** is formed on the n-type cladding layer **21** in the etched second region **R2** by a well-known deposition method such as an electron-beam evaporation method. After at least one of the p-electrode **26** and the n-electrode **27** is formed, heat treatment may be performed by a well-known heat treatment method such as RTA (rapid thermal annealing).

Note that, as an example, the light-emitting element **1** can be used in a state in which it is flip-chip mounted on a base such as a submount and then sealed by a predetermined resin such as a silicone resin or an amorphous fluororesin (e.g., a resin having a lens shape).

<Cross-Sectional Observation and Compositional Analyses of n-type Cladding Layer>

Next, a sample for observing the cross section of the n-type cladding layer **21** is produced, a sample piece having a cross section perpendicular (or substantially perpendicular) to the upper surface of the n-type cladding layer **21** is processed by a focused ion beam (FIB), and the sample piece is observed by a scanning transmission electron microscope (STEM), and the result will be described with reference to the drawings.

The sample was produced by sequentially depositing the n-type cladding layer **21**, the active layer **22**, an AlGaIn layer having a higher AlN mole fraction than the n-type cladding layer **21**, an AlGaIn layer for protecting the sample surfaces, and a protective resin film on the underlying part **10** composed of the sapphire substrate **11** and the AlN layer **12** in accordance with the manufacturing procedure of the n-type cladding layer **21** and the like described above. In the preparation of the sample, the underlying part **10** in which multi-step terraces were appeared on the surface of the AlN layer **12** using the sapphire substrate **11** whose main surface has a miscut angle with respect to the (0001) plane was used. In the preparation of the sample, the film thickness of the n-type cladding layer **21** was set to 2 μm , and the target AlN mole fraction of the n-type cladding layer **21** was set to 58%. In addition, the dose of donor impurity (Si) was controlled so that the donor impurity concentration was about $3 \times 10^{18} \text{ cm}^{-3}$.

FIG. **8** shows the high-angle annular dark-field (HAADF)-STEM image of the cross section of the sample piece. FIG. **8** is a HAADF-STEM image of the entire n-type cladding layer **21** including the upper layer of the AlN layer **12**, the n-type cladding layer **21** and the active layer **22** of the sample piece.

In HAADF-STEM image, contrasts proportional to atomic weights are obtained, and heavy elements are dis-

played brightly. Therefore, as to the first Ga-rich region **21a** and the n-type body region **21b** in the n-type cladding layer **21**, the first Ga-rich region **21a** having lower AlN mole fraction are displayed brighter than the n-type body region **21b**. HAADF-STEM image is more suitable for observing differences in AlN mole fraction than the normal STEM image (bright-field image).

It can be seen from FIG. **8** that a plurality of the first Ga-rich regions **21a**, which are stratiform regions having locally lower AlN mole fraction, are present dispersed in the vertical direction in the n-type cladding layer **21**, and each of the first Ga-rich regions **21a** is extending in a direction inclined with respect to the intersection line between the upper surface of the n-type cladding layer **21** and the first plane on a viewing surface of the HAADF-STEM image (a cross section of the sample piece, corresponding to the first plane). Although each of the first Ga-rich regions **21a** is extending obliquely upward in a linear manner, it is not necessarily extending in a straight line, and the inclination angle with respect to the intersection varies by location even in the same first Ga-rich region **21a**. On the cross section shown in FIG. **8** (corresponding to the first plane), it is also observed that a portion of the first Ga-rich region **21a** intersects with or branches from another first Ga-rich region **21a**.

In the present embodiment, the compositional analyses in the n-type cladding layer **21** of the sample pieces were performed by two types of analysis methods (line analysis of energy dispersive X-ray spectroscopy (cross-sectional TEM-EDX) and CL (cathodoluminescence) method).

In the compositional analysis by EDX method (EDX measurement), an electron beam probe (diameter: about 2 nm) was scanned first longitudinally (vertical direction) and laterally (direction parallel to the second plane) in the entire measurement region covering substantially the entire area of HAADF-STEM image shown in FIG. **8**, and the detection data (X-ray intensity corresponding to the respective Al and Ga compositions) at each probe location distributed in a matrix of 512×512 at about 4 nm intervals in the longitudinal and transverse directions, was obtained.

Next, in order to perform the line-analysis by the EDX measurement to the first Ga-rich region **21a** dispersed in the entire measurement region, six measurement regions A to F having a generally square shape (about 420 nm $\times$ about 420 nm) were set in the entire measurement region as shown in FIG. **9**. FIG. **9** shows a rectangular frame representing respective measurement regions A to F superimposed on HAADF-STEM image of FIG. **8**. Each of the six measurement regions is set so that at least one of the first Ga-rich regions **21a** confirmed on HAADF-STEM image is crossing in each measurement region. The inclination of each measurement region is set for each measurement region so that the extending direction of at least one of the first Ga-rich regions **21a** in the measurement region is perpendicular to the scanning direction of the line analysis. Each inclination of the measurement regions A to F (the angle between the longitudinal direction of the entire measurement region and the longitudinal direction of each measurement region) is approximately equal at about 20°, but strictly speaking, is not necessarily the same. Here, for convenience of explanation, the scanning direction of the line analysis is defined as a longitudinal direction and the direction perpendicular to the scanning direction is defined as a transverse direction in each of the measurement regions A to F of FIG. **9**, apart from the longitudinal and transverse directions of the entire measurement region. The center vertical line shown in each of the measurement regions indicates the scanning direction,

and the center horizontal line indicates the position where said at least one of the first Ga-rich regions **21a** is assumed to be present and is the origin (0 nm) of the scanning position of the line analysis in the compositional analysis described later. Incidentally, the vertical line indicating the scanning direction is marked with an arrow, which points to the direction of the AlN layer **12**. The scanning positions are set on the center vertical line in the longitudinal direction across the above-mentioned origin at intervals of about 5 nm and within the range of 58 to 81 points in total for each measurement region A to F.

In the EDX measurement, since the diameter of the electron beam probe to be irradiated is as small as about 2 nm, the spatial resolution is high, but since the X-rays emitted from respective probe positions are weak, in the line analysis of the present embodiment, respective detection data obtained from a plurality of probe positions aligned in the transverse direction at each scanning position are accumulated to be the detection data at each scanning position. Incidentally, "aligned in the transverse direction" means that the irradiation range of the electron beam probe is overlapped with the horizontal line intersecting the vertical line and extending in the transverse direction at each scanning position.

Therefore, when all the plurality of probe positions aligned in the transverse direction are located within the metastable n-type region of the first Ga-rich region **21a** at a certain scanning position, the accumulated detection data accurately indicates the AlN mole fraction of the metastable n-type region. Similarly, when all the plurality of probe positions aligned in the transverse direction are located within the n-type body region **21b** at a certain scanning position, the accumulated detection data accurately indicates the AlN mole fraction of the n-type body region **21b**.

However, at a certain scanning position, when a part of the plurality of probe positions aligned in the transverse direction, or a part of the probe range (about 2 nm in diameter) of each probe position is located within the metastable neighbor n-type region or the n-type body region **21b** other than the metastable n-type region, due to the extending direction of the metastable n-type region of the first Ga-rich region **21a** being not exactly perpendicular to the scanning direction of the line analysis, or the extending direction of the metastable n-type region of the first Ga-rich region **21a** being not straight such as by bending, or the like, the accumulated detection data indicates an average AlN mole fraction of the plurality of probe positions, and indicates a value higher than the AlN mole fraction of the metastable n-type region.

Similarly, even if most of the plurality of probe positions aligned in the transverse direction are located within the n-type body region **21b** at a certain scanning position, when a part of the plurality of probe positions or a part of the probe range (about 2 nm in diameter) of each probe position is located in a region having locally lower or higher AlN mole fraction caused by the mass-transfer in the n-type body region **21b**, or in a region having locally lower AlN mole fraction other than the n-type body region **21b** (a stratiform region other than the first Ga-rich region **21a**, the metastable n-type region or the metastable neighbor n-type region in the first Ga-rich region **21a**), the accumulated detection data indicates an average AlN mole fraction of the plurality of probe positions, and indicates a value lower or higher than the average AlN mole fraction of the n-type body region **21b** (=target value of the AlN mole fraction of the n-type cladding layer **21**).

FIG. **10A** to FIG. **10F** show the results of compositional analyses in the n-type cladding layer **21** in the six measurement regions A to F shown in FIG. **9** by the linear analysis of the EDX measurement. In the graphs of the compositional analysis results of the measurement regions A to F shown in FIGS. **10A** to **10F**, the horizontal axis indicates the scanning position along the center vertical line of each measurement region, and the vertical axis indicates the measurement results of the AlN mole fraction and the GaN mole fraction. 0 nm of the scanning position of the horizontal axis indicates the position of the center horizontal line shown in each measurement region (where at least one of the first Ga-rich regions **21a** is assumed to be present). The scanning positions are indicated by positive values below the origin (0 nm) (the AlN layer **12** side) and by negative values above the origin (the active layer **22** side), respectively.

In the EDX measurement, as described above, since the X-rays emitted from the probe positions are weak, even if the respective detection data (X-ray intensity of each composition) of the probe positions are accumulated in the transverse direction at each scanning position, the measurement error is generally large. For example, when the calibration is performed based on the AlN mole fraction (100%) of the AlN layer **12** where the AlN mole fraction is predetermined, the measurement error of the detection data at each scanning position is about ± 2 to 3% even near the AlN layer **12** as a reference, and the measurement accuracy further decreases with increasing distance from the AlN layer **12**. Therefore, in the present embodiment, in order to suppress the measurement error at the respective scanning positions to about ± 2 to 3% even in the region apart from the AlN layer **12**, the same sample as the sample piece used for the EDX measurement was used to perform the compositional analysis of Al and Ga in n-type cladding layer **21** by the Rutherford backscattering (RBS) analysis method, and the result obtained by the EDX measurement was calibrated using the RBS analysis result. The AlN mole fraction and the GaN mole fraction of the measurement regions A to F shown in FIGS. **10A** to **10F** show the calibrated results.

From FIG. **10A**, in the measurement region A, the presence of the first Ga-rich region **21a** can be confirmed in the region A1 at the scanning positions of about -140 nm to about -131 nm and in the region A2 at the scanning positions of about -5 nm to about 5 nm. The AlN mole fractions at three scanning positions in region A1 are within 52.4% to 52.8%. The AlN mole fractions at three scanning positions in region A2 are within 50.9% to 51.5% (one of them is within 50% to 51%).

From FIG. **10B**, in the measurement region B, the presence of the first Ga-rich region **21a** can be confirmed in the region B1 at the scanning positions of about -169 nm to about -145 nm and in the region B2 at the scanning positions of about -5 nm to about 5 nm. The AlN mole fractions at six scanning positions in region B1 are within 51.8% to 52.4% (four of them are within 50% to 52%). The AlN mole fractions at three scanning positions in region B2 are within 51.5% and 53.0% (one of them is within 50% to 52%).

From FIG. **10C**, in the measurement region C, the presence of the first Ga-rich region **21a** can be confirmed in the region C1 at the scanning positions of about -53 nm to about -44 nm and in the region C2 at the scanning positions of about 0 nm to about 5 nm. The AlN mole fractions at three scanning positions in region C1 are within 51.2% to 52.6% (two of them are within 50% to 52%). The AlN mole fractions at two scanning positions in region C2 are 51.4% and 52.6%.

From FIG. 10D, in the measurement region D, the presence of the first Ga-rich region **21a** can be confirmed in the region D1 at the scanning positions of about -169 nm to about -97 nm and in the region D2 at the scanning positions of about -5 nm to about 5 nm. The AlN mole fractions at 16 scanning positions in the region D1 are within 50.7% to 52.5% (six of them are within 50% to 51% and 13 of them are within 50% to 52%). The AlN mole fractions at three scanning positions in the region D2 are within 50.8% to 52.6% (two of them are within 50% to 52%).

From FIG. 10E, in the measurement region E, the presence of the first Ga-rich region **21a** can be confirmed in the region E1 at the scanning positions of about -116 nm to about -73 nm and in the region E2 at the scanning positions of about -5 nm to about 5 nm. The AlN mole fractions at ten scanning positions in region E1 are within 51.3% to 52.8% (four of them are within 50% to 52%). The AlN mole fractions at three scanning positions in region E2 are within 50.5% to 51.9% (one of them is within 50% to 51%).

From FIG. 10F, in the measurement region F, the presence of the first Ga-rich region **21a** can be confirmed in the region F1 at the scanning positions of about -160 nm to about -140 nm, in the region F2 at the scanning positions of about -73 nm to about -58 nm, in the region F3 at the scanning positions of about 0 nm to about 15 nm, and in the region F4 at the scanning positions of about 97 nm to about 106 nm. The AlN mole fractions at five scanning positions in the region F1 are within 49.1% to 52.3% (two of them are within 49% to 50%, one of them is within 50% to 52%). The AlN mole fractions at four scanning positions in the region F2 are within 51.4% to 52.7% (two of them are within 50% to 52%). The AlN mole fractions at four scanning positions in the region F3 are within 51.6% to 52.2% (three of them are within 50% to 52%). The AlN mole fractions at three scanning positions in the region F4 are 51.0% to 51.3% (one of them is within 50% to 51%).

As observed above, it is possible to confirm the presence of the metastable n-type regions having the AlN mole fraction of 50% in the first Ga-rich region **21a** in the respective regions A1, A2, B1, B2, C1, C2, D1, D2, E1, E2, and F1 to F4 of the measurement regions A to F, considering the measurement error of about ± 2 to 3% at each scanning position and the possibility that the average AlN mole fraction of the plurality of probe positions aligned in the transverse direction with respect to the first Ga-rich region **21a** is higher than the AlN mole fraction of the metastable n-type region. Furthermore, the first Ga-rich regions **21a** are present respectively in the measurement regions A to C of the upper portion close to the upper surface of the n-type cladding layer **21**, in the measurement region D of the central portion, and in the measurement regions E and F of the lower portion close to the AlN layer **12**, and it can be seen that the first Ga-rich regions **21a** are uniformly dispersed in the n-type cladding layer **21**.

Furthermore, from FIGS. 10A to 10F, it can be confirmed that the AlN mole fractions in the n-type body regions **21b** adjacent to the regions A1, A2, B1, B2, C1, C2, D1, D2, E1, E2, and F1 to F4 of the measurement regions A to F are within the range of about 55% to about 59%. As described above, since the target value of the AlN mole fraction of the n-type cladding layer **21** of the sample used for the EDX measurement is 58%, it can be seen that the FIGS. 10A to 10F accurately represent the AlN mole fraction of the n-type body region **21b**, considering the measurement error of about ± 2 to 3% at each scanning position and the possibility that the average AlN mole fraction of the plurality of probe positions aligned in the transverse direction with respect to

the n-type body region **21b** is higher or lower than the average AlN mole fraction of the n-type body region **21b**.

Next, the measurement results of the AlN mole fractions of the first Ga-rich region **21a** and the n-type body region **21b** in the n-type cladding layer **21** by the CL (cathodoluminescence) method are described. The sample pieces used for the measurement was prepared in the same manner as the sample pieces used for the observation of HAADF-STEM image shown in FIG. 8.

FIG. 11 is a scanning electron microscopy (SEM) image showing a cross section in the n-type cladding layer **21** of the sample piece. The measurement regions (a to f) surrounded by dotted lines in the cross section indicate the incident areas of the electron beam irradiated for the measurement, respectively. The measurement regions a and b are located at about 1800 nm from the upper surface of the AlN layer **12**, the measurement regions c and d are located at about 1000 nm from the upper surface of the AlN layer **12**, and the measurement regions e and f are located at about 400 nm from the upper surface of the AlN layer **12**. Within each measurement region, the electron beam having a beam diameter of 50 nm was moved laterally and irradiated once each at 50 nm intervals for a total of 10 times to measure the CL spectrum at each irradiation.

FIG. 12 shows the first CL spectrum and the second CL spectrum for each measurement region (a to f). The first CL spectrum was obtained by averaging two CL spectra whose wavelength distributions are closer to the short wavelength among the ten CL spectra in each measurement region (a to f) and the second CL spectrum was obtained by averaging two CL spectra whose wavelength distributions are closer to the long wavelength among the ten CL spectra in each measurement region (a to f).

Since the distance between both ends of the ten electron-beam centers in each measurement region (a to f) is 450 nm, both the first Ga-rich region **21a** and the n-type body region **21b** are present in the ten irradiation areas. Since the volume ratio of the first Ga-rich region **21a** to the entire n-type cladding layer **21** is small, the first CL spectrum mainly shows the CL spectrum of the n-type body region **21b**. On the other hand, the second CL spectrum includes the CL spectrum of the first Ga-rich region **21a**. However, since the width in a cross section perpendicular to the extending direction of the first Ga-rich region **21a** is about 20 nm on average, the n-type body region **21b** may be partially included within the irradiation range of the beam diameter 50 nm. Therefore, the second CL spectrum is a composite spectrum of the CL spectrum of the first Ga-rich region **21a** and the CL spectrum of the n-type body region **21b**. However, if the center of each electron beam of the two CL spectra whose wavelength distributions are closer to the long wavelength is located at the widthwise center of the first Ga-rich region **21a**, it is likely that the electron beam of the central portion within the irradiation range will gather in the first Ga-rich region **21a** with lower energy level to excite the first Ga-rich region **21a** exclusively, and it is considered that the second CL spectrum mainly shows the CL spectrum of the first Ga-rich region **21a**.

Here, the reason why the first CL spectrum is the average of the two CL spectra whose wavelength distributions are closer to the short wavelength and the second CL spectrum is the average of the two CL spectra whose wavelength distributions are closer to the long wavelength is as follows. The irradiation positions of the electron beam at each measurement region are set at random, so that the irradiation ranges of one CL spectrum closest to the short wavelength and the one closest to the long wavelength are different for

each measurement region, and the measurement results are largely varied for each measurement region. Also, it may be difficult to sort out one CL spectrum closest to the shortest wavelength and the one closest to the long wavelength. Therefore, it was decided to mechanically select two CL spectra whose wavelength distributions are closer to the short wavelength and the other two CL spectra whose wavelength distributions are closer to the long wavelength, respectively and to take the averages of the respective two CL spectra in order to suppress the variation for each measurement region.

First, the first CL spectrum of each measurement region (a to f) will be discussed. In the measurement region a, a peak of emission wavelength exists near about 262 nm. In the measurement region b, gradual peaks of emission wavelength exist at two locations near about 263 nm and near about 271 nm. In the measurement region c, a peak of emission wavelength exists near about 261 nm. In the measurement region d, a peak of emission wavelength exists near about 262 nm to 263 nm. In the measurement region e, a peak of emission wavelength exists near about 256 nm to 259 nm. In the measurement region f, a peak of emission wavelength exists near about 257 nm to 259 nm.

The peak wavelength of about 256 nm to about 263 nm in each measurement region (a to f) corresponds to about 57% to about 63% in terms of the AlN mole fraction, and the above CL wavelength of the first CL spectrum generally coincides with the average AlN mole fraction X_b (≈target value 58%) of the n-type body region **21b** in consideration of a measurement error of about ±3% in terms of the AlN mole fraction.

In addition, in the first CL spectra of the measurement regions a to f, the long wavelength component longer than a peak wavelength of about 259 nm to about 263 nm is larger than the short wavelength component shorter than the same peak wavelength, and it can be seen that the mass-transfer of Ga occurs within two irradiation ranges corresponding to the first CL spectrum of each measurement region. Furthermore, the peak wavelength of about 271 nm of the first CL spectrum in the measurement region b corresponds to about 53%±3% in terms of the AlN mole fraction, and generally coincides with the CL wavelength from the metastable neighbor n-type region, whose AlN mole fraction is slightly higher than 50%, in the first Ga-rich region **21a**. Therefore, a portion of the two irradiation ranges corresponding to the first CL spectrum of the measurement region b includes the first Ga-rich region **21a** formed by the mass-transfer of Ga.

Next, the second CL spectrum of each measurement region (a to f) will be discussed. In the measurement regions a and b, a peak of emission wavelength exists near about 272 nm. In the measurement region c, gradual peaks of emission wavelength exist at two locations near around about 261 nm and near around 270 nm. In the measurement region d, peaks of emission wavelengths exist at two locations near about 269 nm and near around 271 nm. In the measurement region e, a peak of emission wavelength exists near about 268 nm. In the measurement region f, gradual peaks of emission wavelength exist at two locations near about 260 nm and near about 270 nm, and the overall plateau-like peak region extends in the range of about 260 nm to about 270 nm.

The peak wavelength of about 270 nm to about 272 nm in the measurement regions a to d and f corresponds to about 52% to about 53% in terms of the AlN mole fraction, and generally coincides with the CL wavelength (about 275 nm) corresponding to the metastable n-type region having the AlN mole fraction of 50% present in the first Ga-rich region **21a** in consideration of a measurement error of about ±3% in terms of AlN mole fraction. The second CL spectra of the

measurement regions a to d and f also include the CL wavelength of about 275 nm, which corresponds to the metastable n-type region, with an emission intensity of about 73 to 79% of the peak intensity. However, the peak wavelength of about 270 nm to about 272 nm is about 3 to 5 nm shorter than the CL wavelength (about 275 nm) corresponding to metastable n-type region. These indicate that each of the second CL spectra of the measured regions a to d and f appears as a composite spectrum of the respective CL spectra of the metastable n-type region and the metastable neighbor n-type region having slightly higher AlN mole fraction than the metastable n-type region in the first Ga-rich region **21a** and the CL spectrum of the n-type body region **21b**. Furthermore, in each of the second CL spectra of the measurement regions c, d, and f, a gradual peak or a shoulder (undulation) of the emission wavelength is present around about 260 nm to about 262 nm, and the proportion of the CL spectrum of the n-type body region **21b** in the composite spectrum is larger than that of the measurement regions a and b.

On the other hand, in the second CL spectrum of the measurement region e, the peak wavelength is about 268 nm, which corresponds to about 54% in terms of the AlN mole fraction. The peak wavelength of about 268 nm is about 7 nm shorter than the CL wavelength (about 275 nm) corresponding to the metastable n-type region. The second CL spectrum of the measurement region e includes the CL wavelength of about 275 nm corresponding to the metastable n-type region with an emission intensity of about 46% of the peak intensity. These indicate that the second CL spectrum of the measurement region e appears as the composite spectrum of the respective CL spectra of the metastable n-type region and the metastable neighbor n-type region in the first Ga-rich region **21a** and the CL spectrum of the n-type body region **21b**, as with the measurement regions a to d and f. However, in the second CL spectrum of the measurement region e, the proportion of the CL spectrum of the metastable n-type region to the composite spectrum is smaller than that of the measurement regions a to d and f.

As described above, from the first CL spectra in the respective measurement regions a to f shown in FIG. 12, the AlN mole fraction of the n-type body region **21b** almost coincides with the target value 58% of the AlN mole fraction of the n-type cladding layer **21**. Furthermore, from the second CL spectra in the respective measurement regions a to f, it can be seen that while the first Ga-rich region **21a** includes the metastable n-type region having the AlN mole fraction of 50%, the metastable neighbor n-type region having the AlN mole fraction higher than that of the metastable n-type region is present. Furthermore, the analysis results shown in the first and second CL spectra in the measurement regions a to f shown in FIG. 12 are in good agreement with the analysis results obtained by the EDX measurement shown in FIGS. 10A to 10F, although there are differences in the spatial resolution and the like due to differences in analysis methods.

It should be noted that, from the second CL spectra in the respective measurement regions a to f shown in FIG. 12, the abundance ratio of the metastable n-type region in the first Ga-rich region **21a** tends to change depending on the position in n-type cladding layer **21**. However, since there are many uncertainties, detailed examination is omitted.

Here, even if the abundance ratio of the metastable n-type region is small at the region close to the AlN-layer **12** in the n-type cladding layer **21**, the effect of the present invention is not necessarily reduced. As described above, the carriers (electrons) in the n-type cladding layer **21** are localized in

the first Ga-rich region **21a**, so that the current can flow preferentially through the first Ga-rich region in the n-type cladding layer **21** stably, thereby suppressing variation in the properties of the light-emitting element. However, since the active layer **22**, which is a light-emitting region, is located above the n-type cladding layer **21**, the localization is remarkable in the vicinity of the upper surface of the n-type cladding layer **21** which is in contact with the active layer **22**. Therefore, even if the above localization is insufficient in the area close to the AlN-layer **12** in the n-type cladding layer **21**, it is possible to suppress the variation in the properties of the light-emitting element in the same manner. Furthermore, in the element configuration shown in FIG. 4, since the forward current flows more in the upper layer side than in the lower layer side in the n-type cladding layer **21**, it is considered that there is almost no effect of the insufficient localization in the area close to the AlN layer **12** in the n-type cladding layer **21**.

<CL Spectrum in Well Layer>

FIG. 13 shows the CL spectra of the inclined region BA and the terrace region TA of the well layer **220** measured on a sample having a multi-quantum-well structure in which three layers of the well layer **220** and two layers of the barrier layer **221** are formed as shown in FIG. 5. The sample piece used for the CL measurement were prepared in the same manner as the sample pieces obtained when compositional analyses in the n-type cladding layer **21** were performed by the EDX method and the CL method.

The film thickness of the terrace region TA (the well body region **220b**) in the well layer **220** of the sample piece is 7 to 8 ML, and the film thickness of the inclined region BA (the second Ga-rich region **220a**) is 9 to 10 ML and slightly thicker than the terrace region TA. The target AlN mole fraction of the barrier layer **221** of the sample piece is 80%. Consequently, it is assumed that the metastable AlGaIn in which the AlGaIn composition ratio is an integer ratio of $\text{Al}_3\text{Ga}_1\text{N}_4$ is dominantly formed in the inclined region BA (the third Ga-rich region **221a**) of the barrier layer **221**, and that the AlN mole fraction is 75%.

As shown in FIG. 13, the peak emission wavelength of the CL spectrum at the inclined region BA (the second Ga-rich region **220a**) of the well layer **220** is about 296 nm, which is substantially the same as the emission wavelength of the simulation result shown in FIG. 6. From this, it can be seen that the metastable well region (the metastable AlGaIn in which the AlGaIn composition ratio is $\text{Al}_1\text{Ga}_2\text{N}_3$) is dominantly formed at the second Ga-rich region **220a** in the well layer **220** of the sample piece.

Furthermore, as shown in FIG. 13, the peak emission wavelength of the CL spectrum in the terrace region TA (barrier body region **221b**) of the well layer **220** is about 286 nm, which is about 10 nm shorter than that of the inclined region BA. This is attributed to that the terrace region TA has thinner film thickness and higher AlN mole fraction than those of the inclined region BA.

In the CL spectra shown in FIG. 13, the inclined region BA and the terrace region TA are irradiated with an electron beam having a beam diameter of 50 nm, respectively, so that the respective CL spectra are individually measured. However, in actual LED emission, as described above, the current exciting the emission mainly flows through the inclined regions BA of the well layer **220** and the barrier layer **221**, so that the emission with the peak emission wavelength of about 296 nm is dominant, and the peak of the emission spectrum does not separate into two peaks of about 296 nm and about 286 nm.

Notable in measuring the CL spectrum of the well layer **220** is that, unlike the compositional analysis of the n-type cladding layer **21** by the CL method, the AlN mole fractions of the barrier layer **221**, the n-type cladding layer **21**, and the electron blocking layer which are neighboring the well layer **220** are about 33.3% to 50% higher than the AlN mole fraction (33.3%) of the inclined region BA (the second Ga-rich region **220a**) of the well layer **220**. Therefore, even if the beam diameter of the electron beam irradiated toward the inclined region BA of the well layer **220** is as large as 50 nm, the electrons with high energy at the beam center are concentrated on the inclined region BA of the well layer **220** whose energy level is low, so that the emission wavelength of the inclined region BA of the well layer **220** can be accurately measured.

Second Embodiment

In the first embodiment, when the barrier layer **221** is composed of an AlGaIn based semiconductor having an AlN mole fraction of not 100%, it is described as an example that the AlN mole fraction of the entire barrier layer **221** including the third Ga-rich region **221a** is in the range of 50% to 90%, the AlN mole fraction of the barrier body region **221b** is in the range of 51% to 90%, and the difference in the AlN mole fraction between the third Ga-rich region **221a** and the barrier body region **221b** is 1% or more in order to ensure the effectiveness of the localization of carriers in the third Ga-rich region **221a**.

In the second embodiment, similarly to the first Ga-rich regions **21a** of the n-type cladding layer **21** and the second Ga-rich region **220a** of the well layer **220** in the first embodiment, the third Ga-rich region **221a** of the barrier layer **221** is also preferable to be composed of the first or second metastable AlGaIn. Here, since the AlN mole fraction of the entire barrier layer **221** is in the range of 50% to 90%, in the first metastable AlGaIn applicable to the third Ga-rich region **221a**, the AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_1\text{N}_2$, $\text{Al}_2\text{Ga}_1\text{N}_3$ or $\text{Al}_3\text{Ga}_1\text{N}_4$. Although the second metastable AlGaIn of $\text{Al}_7\text{Ga}_5\text{N}_{12}$ or $\text{Al}_3\text{Ga}_1\text{N}_4$ is also considered to be applicable to the third Ga-rich region **221a**, it is preferable to use a more stable $\text{Al}_3\text{Ga}_1\text{N}_4$ if the second metastable AlGaIn is daringly used. In the second metastable AlGaIn of $\text{Al}_{11}\text{Ga}_1\text{N}_{12}$, since the composition ratio of Al is too high, before easy-to-move Ga enters sites which compose a symmetric arrangement, the quantitatively large amount of Al randomly enters the sites, so that the atomic arrangement of Al and Ga is likely not to be a symmetric arrangement and is close to the random state, and the above stability is reduced. Therefore, it is considered difficult to apply this second metastable AlGaIn to the third Ga-rich region **221a**.

Incidentally, in the simulation result of the emission wavelength of the well layer **220** shown in FIG. 6, the three AlN mole fractions of 66.7%, 75%, and 83.3% are assumed to be used for the third Ga-rich region **221a** of the barrier layer **221**, but these correspond to the AlN mole fractions of the metastable AlGaIns in which AlGaIn composition ratios are $\text{Al}_2\text{Ga}_1\text{N}_3$, $\text{Al}_3\text{Ga}_1\text{N}_4$, and $\text{Al}_5\text{Ga}_1\text{N}_6$.

When the third Ga-rich region **221a** is composed of a metastable AlGaIn of $\text{Al}_1\text{Ga}_1\text{N}_2$, $\text{Al}_2\text{Ga}_1\text{N}_3$, $\text{Al}_3\text{Ga}_1\text{N}_4$, or $\text{Al}_5\text{Ga}_1\text{N}_6$, the AlN mole fraction of the barrier body region **221b** is preferably in a range of 51% to 66%, 68% to 74%, 76% to 82%, or 85% to 90%, depending on the four AlN mole fractions of the third Ga-rich region **221a**. Here, when the third Ga-rich region **221a** is composed of the metastable AlGaIn of $\text{Al}_3\text{Ga}_1\text{N}_4$, it is preferable to set the AlN mole

33

fraction of the barrier body region **221b** not to exceed 90% in order to prevent low stable $\text{Al}_{11}\text{Ga}_1\text{N}_{12}$ being randomly mixed.

In the manufacturing method of the third Ga-rich region **221a** and the barrier body region **221b** in the barrier layer **221**, as described above, the barrier layer **221** is grown under the growth conditions in which multi-step terraces easily appear using the AlN mole fraction set for the barrier body region **221b** as a target value in the same manner as the n-type cladding layer **21**.

When the first metastable AlGa_N of $\text{Al}_1\text{Ga}_1\text{N}_2$ is grown in the third Ga-rich region **221a**, the target value Xd of the AlN mole fraction of the barrier layer **221** is set within the range of 51% to 66% as well as the target value Xa of the AlN mole fraction of the n-type cladding layer **21**. Similarly, when the first metastable AlGa_N of $\text{Al}_2\text{Ga}_1\text{N}_3$ is grown in the third Ga-rich region **221a**, the target value Xd of the AlN mole fraction of the barrier layer **221** is set within the range of 68% to 74%, and when the second metastable AlGa_N of $\text{Al}_3\text{Ga}_1\text{N}_4$ is grown in the third Ga-rich region **221a**, the target value Xd of the AlN mole fraction of the barrier layer **221** is set within the range of 76% to 82%, and when the first metastable AlGa_N of $\text{Al}_5\text{Ga}_1\text{N}_6$ is grown in the third Ga-rich region **221a**, the target value Xd of the AlN mole fraction of the barrier layer **221** is set within the range of 85% to 90%.

Therefore, the target value Xd of the AlN mole fraction of the barrier layer **221** is set within a range of 1% or higher than the AlN mole fraction of the metastable AlGa_N formed in the third Ga-rich region **221a** (target metastable AlGa_N) and less than the AlN mole fraction of the metastable AlGa_N which is nearest to and larger than that of the target metastable AlGa_N. Therefore, as with the first Ga-rich region **21a** of the n-type cladding layer **21**, the target metastable AlGa_N can be stably formed in the third Ga-rich region **221a**, and 1% or more of the AlN mole fraction difference between the third Ga-rich region **221a** and the barrier body region **221b** is ensured, and the carriers in the barrier layer **221** are localized in the third Ga-rich region **221a** having a smaller band gap energy than the barrier body region **221b**.

Because the third Ga-rich region **221a** is composed of highly stable metastable AlGa_N, variation in the mole fraction of mixed crystals caused by the drift of the crystal growth apparatus or the like is suppressed, and the third Ga-rich region **221a** in which carrier localization occurs in the barrier layer **221** is stably formed with the AlN mole fraction corresponding to the metastable AlGa_N used. Consequently, the current can stably flow preferentially through the third Ga-rich region **221a** even in the barrier layer **221** as in the n-type cladding layer **21**, and the variation in the properties of the light-emitting element **1** can be suppressed.

Other Embodiments

Modifications of the first and second embodiments will be described below.

- (1) In the first and second embodiments, the active layer **22** was assumed to be composed of the multi-quantum-well structure in which the well layer **220** and the barrier layer **221** are stacked alternately, where the well layer **220** includes two or more layers composed of an AlGa_N-based semiconductor and the barrier layer **221** includes one or more layers composed of an AlGa_N-based semiconductor or an AlN-based semiconductor. However, the active layer **22** may be configured with a single quantum-well structure having only one layer of the well layer **220**, and it may be configured not to include the barrier layer **221** (quantum barrier layer). It

34

is obvious that the advantages of the n-type cladding layer **21** adopted in the above-described embodiments can be obtained in the same manner for such a single-quantum-well structure.

- (2) In the above embodiments, as an example of the growth condition of the n-type cladding layer **21**, the supply amount and the flow rate of the source gases and the carrier gas used in the MOVPE method are set in accordance with the average AlN mole fraction of the entire n-type AlGa_N layer constituting the n-type cladding layer **21**. That is, when the average AlN mole fraction of the entire n-type cladding layer **21** is set to a constant value vertically, it is assumed that the supply amount and the flow rate of the source gases and the like are controlled to be constant. However, the supply amount and the flow rate of the source gases and the like are not necessarily controlled to be constant.
- (3) In the above embodiments, the planarly-viewed shapes of the first region **R1** and the p-electrode **26** are exemplarily a comb-like shape, but the planarly-viewed shapes are not limited to the comb-like shape. In addition, a plurality of the first regions **R1** may be present, and each of them may be formed in a planarly-viewed shape surrounded by one second region **R2**.
- (4) In the above embodiments, the case using the underlying part **10**, in which the sapphire substrate **11** in which the main surface has a miscut angle with respect to the (0001) surface is used and the multi-step terraces appear on the surface of the AlN layer **12**, is exemplified. However, the magnitude of the miscut angle and the direction in which the miscut angle is provided (specifically, the direction in which the (0001) surface is inclined, for example, the m-axis direction, the a-axis direction, and the like) may be arbitrarily determined as long as the multi-step terraces appear on the surface of the AlN layer **12** and the growth start points of the first Ga-rich region **21a** are formed.
- (5) In the above embodiments, as illustrated in FIG. 1, the light-emitting element **1** having the underlying part **10** including the sapphire substrate **11** is illustrated as the light-emitting element **1**, but the sapphire substrate **11** (or a part or all of layers included in the underlying part **10**) may be removed by lift-off or the like. Furthermore, the substrate constituting the underlying part **10** is not limited to the sapphire substrate.

INDUSTRIAL APPLICABILITY

The present invention is available to a nitride semiconductor ultraviolet light-emitting element comprising a light-emitting element structure part with an n-type layer, an active layer, and a p-type layer stacked vertically, which are made of AlGa_N-based semiconductors with wurtzite structure.

DESCRIPTION OF SYMBOLS

- 1** Nitride semiconductor ultraviolet light-emitting element
- 10** underlying part
- 11** sapphire substrate
- 11a** main surface of sapphire substrate
- 12** AlN layer
- 20** light-emitting element structure part
- 21** n-type cladding layer (n-type layer)
- 21a** first Ga-rich region (n-type layer)
- 21b** n-type body region (n-type layer)
- 22** active layer

35

220 well layer
 220a second Ga-rich region
 220b well body region
 221 barrier layer
 221a third Ga-rich region
 221b barrier body region
 23 electron blocking layer (p-type layer)
 24 p-type cladding layer (p-type layer)
 25 p-type contact layer (p-type layer)
 26 p-electrode
 27 n-electrode
 100 substrate
 101 AlGaIn-based semiconductor layer
 102 template
 103 n-type AlGaIn-based semiconductor layer
 104 active layer
 105 p-type AlGaIn-based semiconductor layer
 106 p-type contact layer
 107 n-electrode
 108 p-electrode
 BL boundary line between first region and second region
 BA boundary region (inclined region)
 R1 first region
 R2 second region
 T terrace
 TA terrace region

The invention claimed is:

1. A nitride semiconductor ultraviolet light-emitting element comprising:

a light-emitting element structure part in which an n-type layer, an active layer, and a p-type layer made of an AlGaIn-based semiconductor of wurtzite structure are stacked vertically, wherein

the n-type layer is composed of an n-type AlGaIn-based semiconductor,

the active layer disposed between the n-type layer and the p-type layer has a quantum-well structure having one or more well layers composed of an AlGaIn-based semiconductor, the p-type layer is composed of a p-type AlGaIn-based semiconductor,

each semiconductor layer in the n-type layer and the active layer is an epitaxially grown layer having a surface on which multi-step terraces parallel to a (0001) plane are formed,

the n-type layer has a plurality of first Ga-rich regions, the plurality of first Ga-rich regions being stratiform regions uniformly distributed in the n-type layer with locally lower AlN mole fraction and including n-type AlGaIn regions in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_1\text{N}_2$,

each extending direction of the first Ga-rich regions on a first plane perpendicular to an upper surface of the n-type layer is inclined with respect to an intersection line between the upper surface and the first plane of the n-type layer,

boundary region parts between adjacent terraces of the multi-step terraces of the well layer have a second Ga-rich region with locally lower AlN mole fraction in the same well layer, and

the second Ga-rich region includes an AlGaIn region in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_1\text{N}_3$.

2. The nitride semiconductor ultraviolet light-emitting element according to claim 1, wherein an AlN mole fraction of an n-type body region outside the first Ga-rich region in the n-type layer is within a range of 54% to 66%.

36

3. The nitride semiconductor ultraviolet light-emitting element according to claim 1, wherein an AlN mole fraction other than the boundary region parts in the well layer is within a range of 33.4% to 37%.

4. The nitride semiconductor ultraviolet light-emitting element according to claim 1, wherein the active layer has a multi-quantum-well structure including two or more well layers, and a barrier layer composed of AlGaIn-based semiconductor is present between two of the well layers.

5. The nitride semiconductor ultraviolet light-emitting element according to claim 4, wherein the barrier layer is composed of an AlGaIn-based semiconductor, and

boundary region parts between adjacent terraces of the multi-step terraces of the barrier layer at least on the most p-type layer side among the barrier layers located between two of the well layers has a third Ga-rich region with a locally lower AlN mole fraction within the same barrier layer.

6. The nitride semiconductor ultraviolet light-emitting element according to claim 5, wherein an AlGaIn region in which AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_1\text{N}_2$, $\text{Al}_2\text{Ga}_1\text{N}_3$, $\text{Al}_3\text{Ga}_1\text{N}_4$ or $\text{Al}_5\text{Ga}_1\text{N}_6$ exists in the third Ga-rich region of the barrier layer.

7. The nitride semiconductor ultraviolet light-emitting element according to claim 1, comprising an underlying part containing a sapphire substrate, wherein

the sapphire substrate has a main surface inclined by a predetermined angle with respect to the (0001) plane, the light-emitting element structure part is formed above the main surface, and

each semiconductor layer at least from the main surface of the sapphire substrate to the surface of the active layer is an epitaxially grown layer having a surface on which multi-step terraces parallel to the (0001) plane are formed.

8. A method for manufacturing a nitride semiconductor ultraviolet light-emitting element comprising a light-emitting element structure part in which an n-type layer, an active layer, and a p-type layer made of an AlGaIn-based semiconductor of wurtzite structure are stacked vertically, the method comprising:

a first operation of epitaxially growing the n-type layer of an n-type AlGaIn-based semiconductor on an underlying part including a sapphire substrate having a main surface inclined by a predetermined angle with respect to a (0001) plane, and making multi-step terraces parallel to the (0001) plane appear on a surface of the n-type layer,

a second operation of epitaxially growing the active layer of a quantum-well structure having one or more well layers composed of an AlGaIn-based semiconductor on the n-type layer, and making multi-step terraces parallel to the (0001) plane appear on a surface of the well layer, and

a third operation of forming the p-type layer of a p-type AlGaIn-based semiconductor on the active layer by epitaxial growth, wherein

in the first operation, a plurality of first Ga-rich regions are grown so as to extend obliquely upward, the plurality of first Ga-rich regions being stratiform regions uniformly distributed in the n-type layer with locally lower AlN mole fraction and including n-type AlGaIn regions in which an AlGaIn composition ratio is an integer ratio of $\text{Al}_1\text{Ga}_1\text{N}_2$, and

37

in the second operation, a second Ga-rich region with locally lower AlN mole fraction in the same well layer is formed in boundary region parts between adjacent terraces of the multi-step terraces of the well layer, and an AlGa_N region in which an AlGa_N composition ratio is an integer ratio of Al₁Ga₁N₃ is grown in the second Ga-rich region.

9. The method for manufacturing a nitride semiconductor ultraviolet light-emitting element according to claim 8, comprising

in the first operation, setting a target AlN mole fraction of the n-type layer within a range of 54% to 66% and growing the n-type AlGa_N region in which AlGa_N composition ratio is an integer ratio of Al₁Ga₁N₂ in the first Ga-rich regions.

10. The method for manufacturing a nitride semiconductor ultraviolet light-emitting element according to claim 8, comprising

in the second operation, setting a target AlN mole fraction of the well layer within a range of 33.4% to 37% and growing AlGa_N region in which AlGa_N composition ratio is an integer ratio of Al₁Ga₁N₃ in the second Ga-rich region.

11. The method for manufacturing a nitride semiconductor ultraviolet light-emitting element according to claim 8, comprising

in the second operation, stacking a well layer composed of an AlGa_N-based semiconductor and a barrier layer composed of an AlGa_N-based semiconductor alternately by epitaxial growth, and forming the active layer of the multi-quantum-well structure including two or more well layers, in which multi-step terraces parallel to the (0001) plane appear on each surface of the barrier layer and the well layer.

38

12. The method for manufacturing a nitride semiconductor ultraviolet light-emitting element according to claim 11, comprising

when forming the barrier layer composed of an AlGa_N-based semiconductor in the second operation, forming a third Ga-rich region having a locally lower AlN mole fraction within the same barrier layer in boundary region parts between the terraces of at least the most p-type layer side among the barrier layers located between two of the well layers.

13. The method for manufacturing a nitride semiconductor ultraviolet light-emitting element according to claim 12, comprising:

in the second operation,

1. growing an AlGa_N region in which AlGa_N composition ratio is an integer ratio of Al₁Ga₁N₂ in the third Ga-rich region by setting a target AlN mole fraction of the barrier layer within a range of 51% to 66%, or
2. growing an AlGa_N region in which AlGa_N composition ratio is an integer ratio of Al₂Ga₁N₃ in the third Ga-rich region by setting a target AlN mole fraction of the barrier layer within a range of 68% and 74%, or
3. growing an AlGa_N region in which AlGa_N composition ratio is an integer ratio of Al₃Ga₁N₄ in the third Ga-rich region by setting a target AlN mole fraction of the barrier layer within a range of 76% to 82%, or
4. growing an AlGa_N region in which AlGa_N composition ratio is an integer ratio of Al₅Ga₁N₆ in the third Ga-rich region by setting a target AlN mole fraction of the barrier layer within a range of 85% to 90%.

* * * * *